Databases II Course Project
OSDP Access Controller Proof of Concept Database

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Abstract

This project's all about getting a proof-of-concept OSDP access controller off the ground, with MongoDB right at its core. I used real hardware - a Raspberry Pi 3B, my own STM32 Blue Pill bridge, RS-485 communication, relay-driving hardware, a Flask backend, and a React operator panel - all tied together through a MongoDB database named osdp_access and live Socket.IO updates. Every important piece of data, from users and credentials to schedules, hardware events, access decisions, and diagnostics, ends up in MongoDB.

Forget textbook CRUD demos. This grew out of actual firmware development, where I needed a flexible controller for real reader hardware. The database, hardware, firmware, and both software layers all evolved together and kept influencing each other as the project moved forward.

The report breaks the project into three main blocks. First, it gives context: what access controllers do, what OSDP is about, and how the prototype looks and works. Second, it dives into the software: STM32 firmware details, backend API flows, frontend logic, and deployment on the Raspberry Pi. Third, it zooms in on MongoDB: the collections, document structure, indexing choices, when to embed versus reference, example queries, plus backup and restore. I also included annotated code and command snippets wherever the implementation details mattered.

Keywords: MongoDB, NoSQL, access control, OSDP, RS-485, STM32 Blue Pill, Raspberry Pi, PyMongo, audit logging, embedded systems

1. Introduction

1.1 Background and Personal Motivation

This project came right out of what I do day to day. As a firmware engineer, I needed a real tool for testing access-control readers, not a sterile software mockup. Instead of separating the course project from actual engineering work, I used it to build something practical while learning more about MongoDB and NoSQL data modeling at the same time.

It turned into a multi-track effort that included:

1. building the physical prototype on a breadboard and inside an enclosure,
2. wiring and debugging the electronics,
3. developing and testing the STM32 bridge firmware,
4. designing the MongoDB schema,
5. implementing the Flask backend and React frontend,
6. deploying the final system on a Raspberry Pi.

This was not just code. A lot of time went into soldering, drilling, wiring, flashing firmware, and hunting communication bugs. Each hardware step changed what I needed from the database, so real engineering decisions kept shaping what had to be tracked and how the data model had to work.

1.2 Problem Statement

Testing a reader is not enough by itself. To test it properly, an engineer also needs the controller side of the system. A useful controller environment has to:

1. communicate over the correct field bus,
2. send and receive real protocol messages,
3. issue commands such as LED, buzzer, relay, identification, capability, and secure-channel requests,
4. record card and keypad events,
5. keep audit logs of access attempts,
6. provide a user interface for configuration and debugging,
7. persist both administrative records and high-volume operational logs.

Some of that data is stable, like users and credentials. Some of it changes constantly, like reader state and hardware event logs. Diagnostic logs are especially useful during debugging, but they do not fit neatly into a rigid relational structure without making the schema awkward and overly verbose.

That is why I used MongoDB as the persistence layer for this proof-of-concept controller.

1.3 Project Scope

The project includes:

1. a MongoDB database for access-control data and logs,
2. a Python backend built with Flask and Flask-SocketIO,
3. a React frontend for operators,
4. a thread-safe serial bridge client in Python,
5. custom STM32 firmware,
6. hardware deployment on Raspberry Pi,
7. backup and restore scripts,
8. firmware update support for the STM32 bridge,
9. source code for the bridge firmware and USB bootloader.

The project does not include:

1. enterprise identity management or LDAP integration,
2. production-grade clustered MongoDB deployment,
3. multiple site management or sharding,
4. biometric template storage,
5. cloud synchronization,
6. the proprietary reader firmware used during testing.

1.4 Objectives

The main objectives of this project are:

1. design a MongoDB schema that fits controller logic and access patterns,

2. handle both stable administrative data and volatile field data,
3. fully audit card and PIN usage,
4. track reader state separately from event history,
5. build a practical controller for OSDP reader development,
6. tie real hardware to a meaningful NoSQL design,
7. demonstrate deployment, backup, restore, and field testing,
8. produce a report that explains both the database and the complete system around it.

1.5 Why This Project Belongs in Databases II

Databases II allows MongoDB and other NoSQL topics. This project fits that well because MongoDB is not just tacked on at the end. It sits at the center of the system and models:

1. users and credentials,
2. schedules and access policy,
3. reader state snapshots,
4. raw hardware events,
5. access decisions,
6. system diagnostics,
7. operational backup and restore.

The design is driven by real runtime needs rather than by made-up sample data, which makes it a strong NoSQL case study.

2. Access Control and OSDP Context

2.1 What an Access Controller Does

In access control, the reader sits at the door waiting for cards or PINs, but the controller is the part that decides who gets in, triggers locks and buzzers, logs every attempt, and keeps track of valid users, rules, and schedules.

Conceptually, the controller has four main responsibilities:

1. communicate with one or more field devices,
2. know which users and credentials are valid,
3. apply business rules such as time schedules and allowed doors,
4. generate output actions such as unlock relay, LED patterns, and buzzer feedback.

That is exactly the controller role implemented in this proof of concept.

2.2 Why a Reader Test Controller Is Valuable

When you are developing readers, protocol sniffers or basic desktop tools are not enough. You need a controller that behaves like the real thing. That means:

1. the reader can be polled,

2. secure channel can be attempted,
3. LED and buzzer commands can be observed,
4. keypad and card events can be captured,
5. reader identification and capabilities can be requested,
6. communication problems can be logged and analyzed.

This setup was important for the firmware side of the work because it gave me a reusable controller environment for reader development, not just an academic example.

2.3 What Is OSDP

OSDP, or Open Supervised Device Protocol, is an access-control standard from the Security Industry Association. It exists to make controller-reader communication open, interoperable, and more secure through features such as AES-128 secure channel. Compared with older interfaces, OSDP is bidirectional, lets you supervise devices, and supports richer control over a simple RS-485 link.

In practical terms, OSDP gives this project several advantages over older one-way interfaces:

1. bidirectional communication,
2. richer device control,
3. reader supervision and status reporting,
4. better interoperability,
5. secure channel support,
6. simpler field wiring through RS-485 multidrop communication.

Industry-facing OSDP white paper cover used as a reference visual during documentation

I used Farpointe Data reference material as supporting documentation while writing the report so the explanation stayed close to current industry presentation, even though those visuals were not part of the implementation itself.

2.4 OSDP Terms Used in This Project

Several OSDP terms appear throughout this project:

1. CP: Control Panel. In this project, that means the Raspberry Pi plus the STM32 bridge.
2. PD: Peripheral Device. In this project, that is the reader being tested.
3. RS-485: The field bus used for communication.
4. Secure Channel: The encrypted and authenticated OSDP mode.
5. SCBK: The Secure Channel Base Key.
6. COMSET: The command used to change communication settings.
7. PDID and PDCAP: The reader's identification and capability responses.

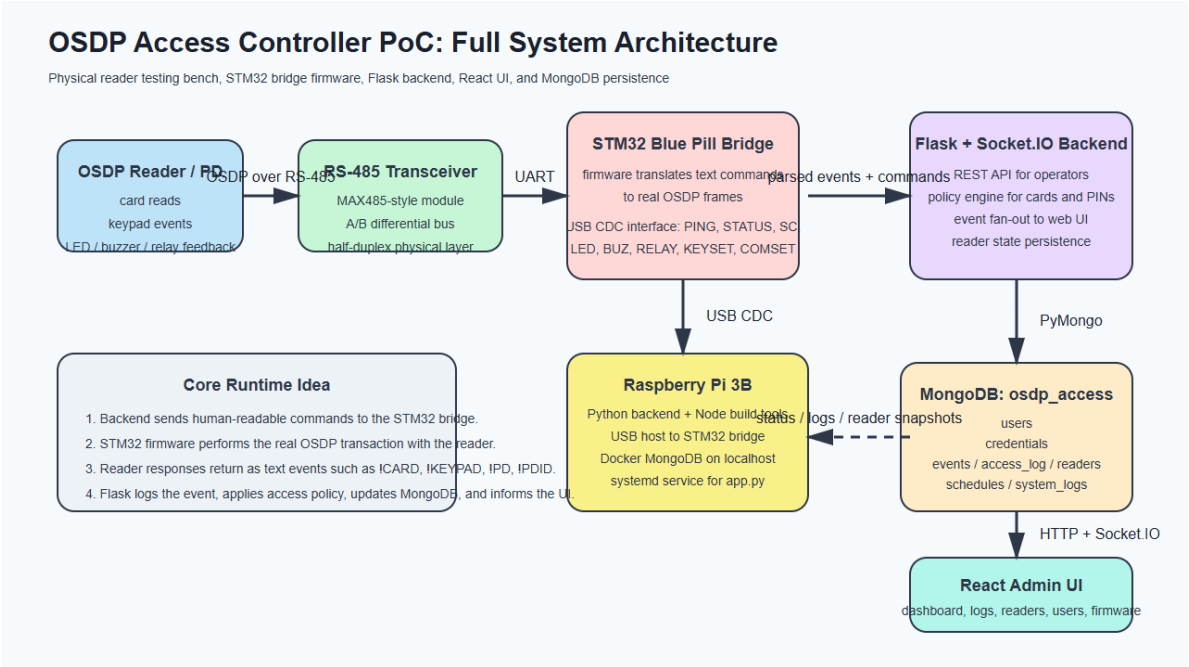
2.5 Why OSDP Instead of Legacy Wiegand-Style Thinking

OSDP matters because it is genuinely interactive. A controller can ask the reader for identity, capabilities, or status, can trigger outputs, can bring up secure channel, and can supervise the link. That makes it much more useful for testing advanced reader behavior than a simple one-way credential interface.

3. Full System Overview

3.1 System Architecture

The easiest way to understand the project is to picture a full embedded access-control stack. The following figure shows the main runtime architecture.



Full system architecture

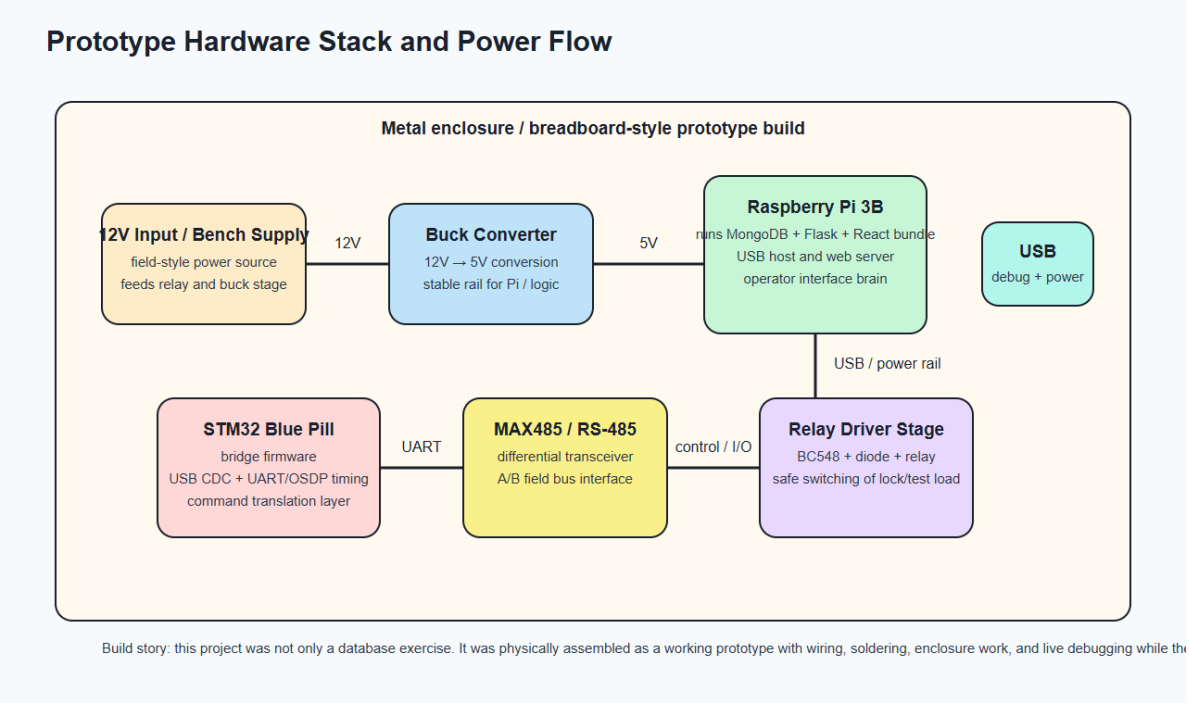
The architecture has five major layers:

1. the reader under test,
2. the RS-485 electrical interface,
3. the STM32 Blue Pill bridge firmware,
4. the Raspberry Pi application stack,
5. the MongoDB database and browser UI.

Python does not generate raw OSDP frames directly. The STM32 bridge handles the field protocol and exposes a simple USB CDC text interface. Flask sends commands such as PING, STATUS, SC 0, LED 0 ..., and BUZ 0 ..., and the bridge sends back text events such as !CARD, !KEYPAD, !PD, and !PDID.

3.2 Hardware Stack and Prototype Build

The prototype uses real hardware: a Raspberry Pi 3B, an STM32 Blue Pill, an RS-485 transceiver, power hardware, and a relay-driving stage.



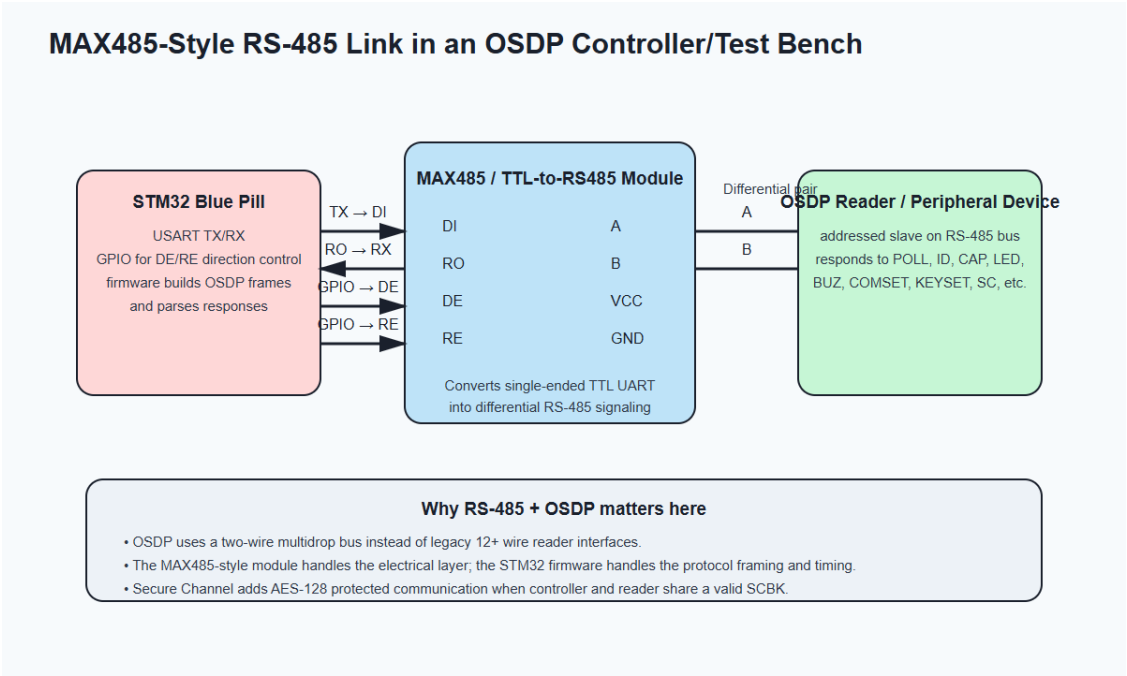
Prototype hardware stack

The physical build process mattered because the software was built against the real prototype, not against a simulation. That changed a lot of design decisions:

1. the backend had to survive USB reconnects,
2. the database had to persist unstable field events for debugging,
3. the Pi deployment script had to be practical on real ARM hardware,
4. power and signal wiring needed to be reflected in the system explanation.

3.3 RS-485 and MAX485-Style Interface

RS-485 is the field link in this setup. A MAX485-style transceiver converts STM32 UART traffic into the differential A/B bus used by the reader.



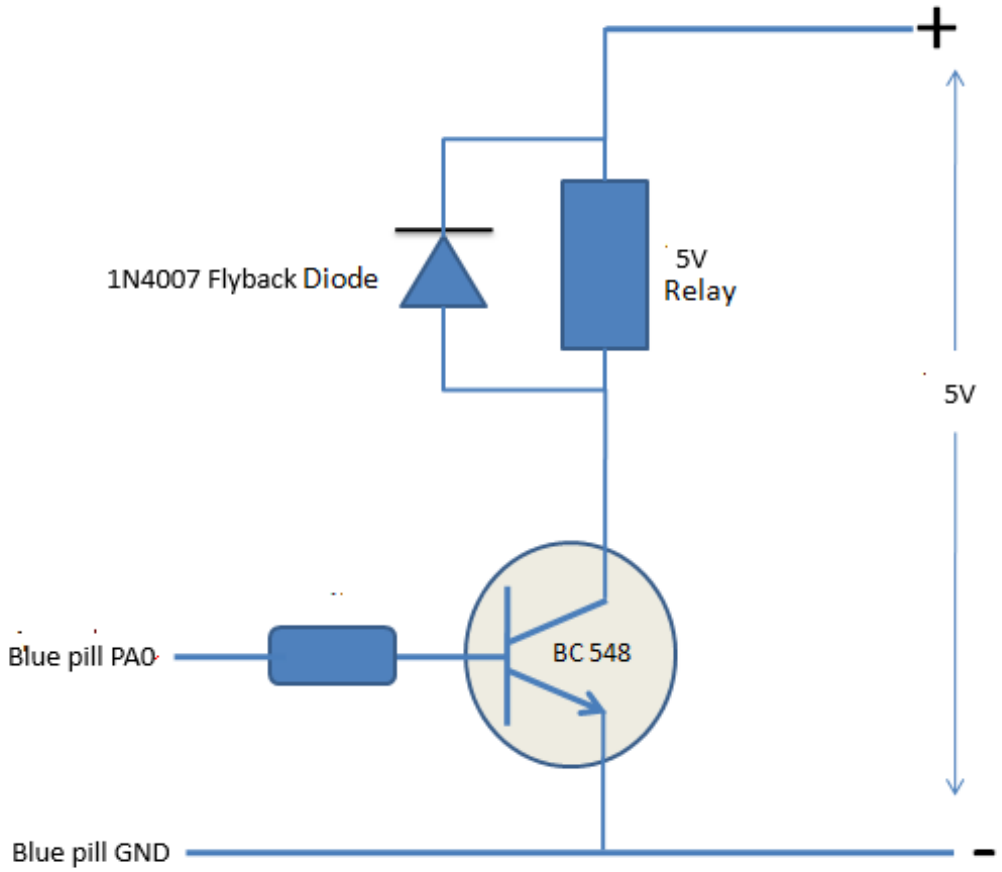
MAX485 style RS-485 path

That layer matters because it is where board-level electrical communication becomes real field communication. The STM32 firmware is responsible for:

1. controlling direction on the half-duplex bus,
2. formatting OSDP frames at the protocol level,
3. timing requests and responses correctly,
4. translating hardware results back into textual events for the Raspberry Pi.

3.4 Relay Driver Stage with BC548

The relay path uses a transistor stage because a microcontroller pin cannot drive a relay coil directly. The following figure shows the concept relevant to this project.



BC548 relay driver concept

The BC548 transistor handles the actual drive current, and the flyback diode protects the rest of the circuit from the relay's turn-off spike.

3.5 Visual Reference Notes

I prepared the report using a mix of original diagrams and representative hardware reference photos. The relay, RS-485 path, prototype stack, and overall system architecture diagrams are original. The board and module photos are included only as representative visuals for the types of hardware used in the prototype. The following source pages were used for those reference images:

1. STM32 Blue Pill style board image reference: Wikimedia Commons board photo
2. Buck converter module image reference: Wikimedia Commons LM2596 module photo
3. Raspberry Pi family board image reference: Wikimedia Commons Raspberry Pi board photo
4. OSDP explainer and white paper reference: Farpointe Data OSDP page

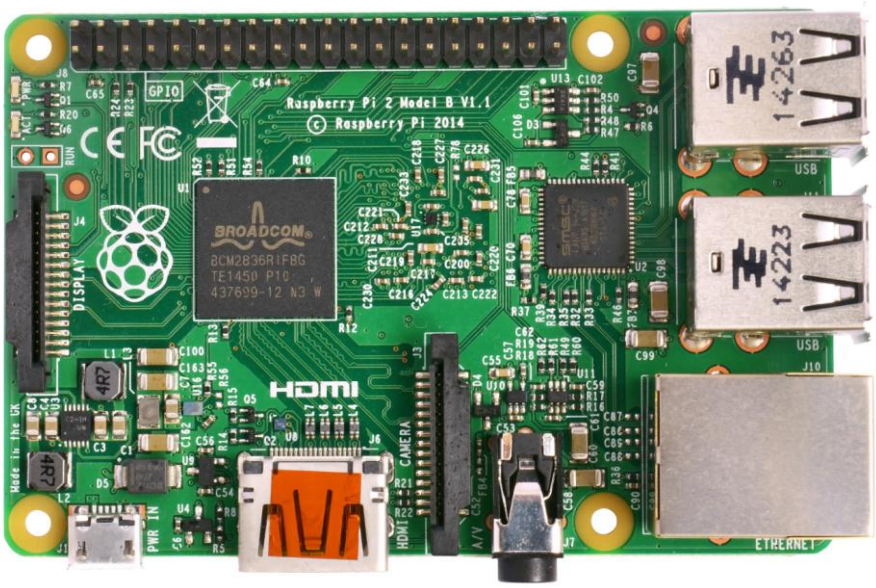
The deployed system itself uses a Raspberry Pi 3B. The Raspberry Pi reference image in the report is representative of the board family, not a photo of the exact unit used in the prototype.

4. Hardware and Firmware Design

4.1 Raspberry Pi 3B Role

The Raspberry Pi is the main brain of the prototype. It runs:

1. the Python backend,
2. the built React frontend,
3. the MongoDB database in Docker,
4. the systemd service used for automatic startup.



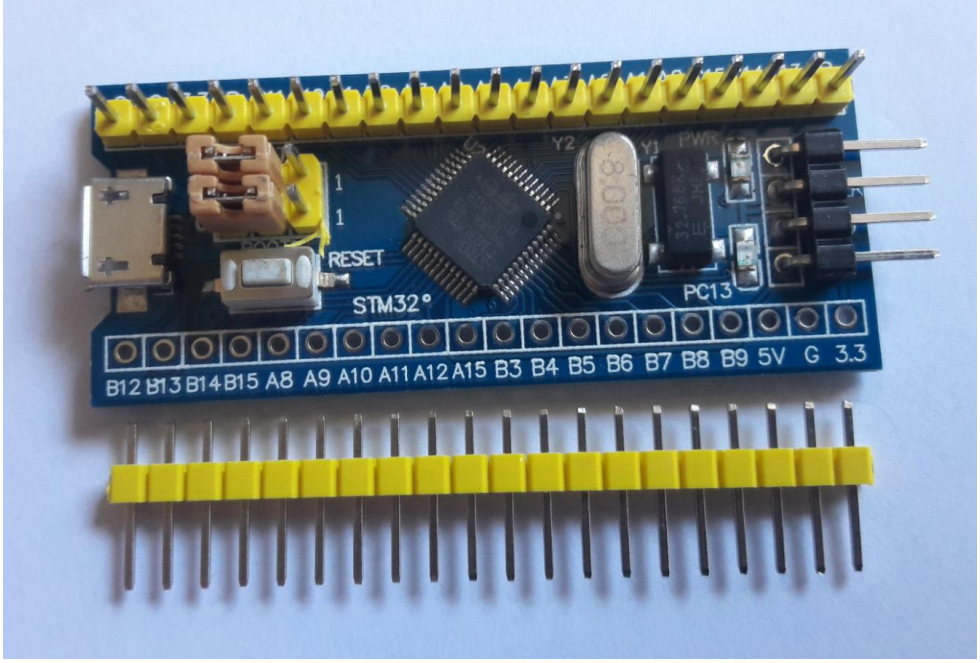
Representative Raspberry Pi family board used as a visual reference for the controller host

It was a practical choice because it is small, inexpensive, Linux-based, and realistic for a field-style controller proof of concept.

4.2 STM32 Blue Pill Role

The STM32 Blue Pill acts as the bridge between the Raspberry Pi and the reader. Its job is to translate between two different worlds:

1. the Raspberry Pi speaks high-level ASCII commands over USB CDC,
2. the reader speaks OSDP over RS-485.



Representative STM32 Blue Pill style development board used as a bridge hardware reference

That keeps the Python backend much simpler, because low-level frame timing and bus direction switching stay inside the embedded firmware.

4.3 What the Repository Firmware and Python Code Reveal About Bridge Behavior

The STM32 bridge firmware lives in osdp-controller/src/main.cpp and osdp-controller/src/osdp_cp.cpp, while the USB bootloader lives in bootloader/src/main.cpp. The Python client in bridge.py gives the clearest host-side picture of how the backend talks to the MCU. The backend sends commands such as:

1. PING
2. STATUS
3. ID <idx>
4. CAP <idx>
5. LSTAT <idx>
6. SC <idx>
7. KEYSET <idx> <key>
8. LED <idx> ...
9. BUZ <idx> ...
10. RELAY <idx> ...

The bridge then emits text events such as:

1. !CARD
2. !KEYPAD
3. !STATE

4. !PD
5. !PDID
6. !PDCAP
7. !LSTAT
8. !NAK
9. !COM

That shows the firmware has three major responsibilities:

1. implement transport between USB CDC and RS-485,
2. translate controller requests into OSDP operations,
3. normalize reader responses into line-oriented textual events.

4.4 Firmware Update Path

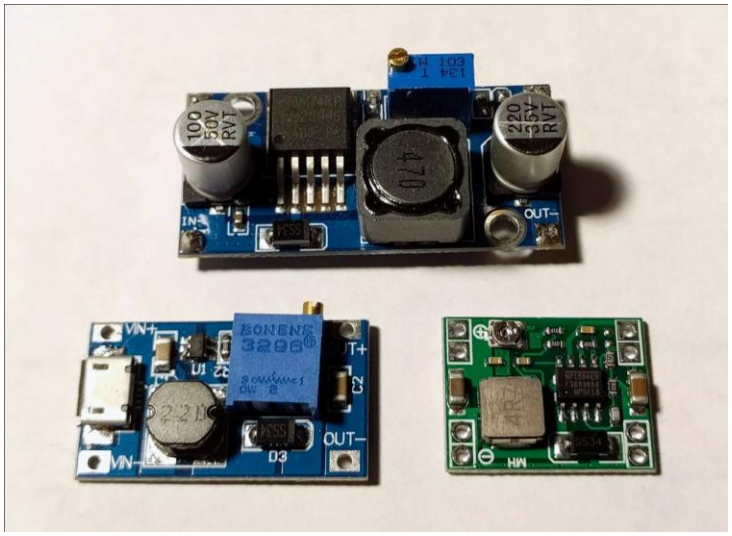
Firmware updates are handled through flasher.py. The flashing logic shows that the Blue Pill uses a custom bootloader workflow:

1. the application firmware can be asked to reboot into bootloader mode,
2. the bootloader enumerates on a different USB PID,
3. the host uploads firmware in small chunks,
4. the bootloader erases flash, writes the new image, verifies CRC-32, and boots the application.

That makes the prototype much easier to improve because the controller can be updated without extra dedicated programming hardware every time.

4.5 Power Path and Buck Converter

The power path matters because the prototype combines different voltage domains. A field-style 12V supply is common in access control, but the Raspberry Pi and the rest of the logic need 5V. The buck converter handles that step-down efficiently and keeps the system powered reliably.



Representative LM2596-style buck converter module used as a power-path reference image

4.6 Relay Path and Test-Bench Outputs

The relay stage matters because access-control systems ultimately have to switch something in the physical world, whether that is a lock, strike, buzzer, or test load. Even on a bench, it is useful to test real outputs instead of treating the controller as a passive logger.

4.7 Engineering Reality of the Prototype

This part of the project was never just a database exercise. The hardware and database kept evolving together during implementation. While building the enclosure, wiring the boards, and dealing with USB, power, and serial problems, the design also had to answer:

1. what should be logged permanently,
2. what belongs in raw event history versus current reader state,
3. how to record denied-access reasons clearly,
4. how to back up the system for redeployment,
5. how to expose the system to a frontend without losing field context.

5. Backend and Frontend Architecture

5.1 Backend Overview

The backend lives in app.py. It uses Flask for REST endpoints and Flask-SocketIO for real-time updates to the frontend. The backend has several responsibilities:

1. connect to MongoDB,
2. authenticate web-panel operators through session-based login,
3. connect to the serial bridge,
4. receive events from the STM32,
5. persist important events in MongoDB,
6. apply access policy for cards and PINs,
7. issue reader feedback commands,
8. enforce admin-only write operations while allowing a read-only demo session,
9. serve the built frontend.

The code exposes more than forty API routes covering users, credentials, schedules, events, system logs, reader commands, firmware actions, and panel authentication.

5.2 Event Processing Strategy

One of the most important backend design decisions is the event queue between the serial thread and the business-logic path. The bridge thread pushes events into a queue, and a worker thread processes them. That keeps the serial loop fast even when field traffic becomes bursty.

5.3 Access Decision Path

When a card arrives, the backend follows this logic:

1. persist the raw card event in events,
2. check whether enrollment mode is active,
3. if not enrolling, look up the card in credentials,
4. load the referenced user from users,
5. evaluate active state, allowed readers, and schedule,
6. write a result to access_log,
7. command relay, LED, and buzzer feedback.

The PIN workflow is similar, except the digits first pass through a small keypad buffer before the final credential lookup.

5.4 Reader State Tracking

The backend keeps historical event logging separate from current reader state. Instead of rebuilding the latest state from the raw event stream every time, the application updates the readers collection through upserts. That keeps the dashboard and diagnostics pages fast.

5.5 Frontend Overview

The frontend is built with React and Vite. frontend/src/App.jsx shows the top-level navigation and the real-time socket listeners. The interface includes pages for:

1. Dashboard
2. Readers
3. Users
4. Enrollment
5. Schedules
6. Events
7. Access Log
8. Reader Config
9. Comms Monitor
10. System Logs
11. Terminal
12. Firmware Update

The panel starts with a login screen and two seeded web accounts. admin / osdp has full control, while demo / db2 is limited to a read-only course-demonstration view. The frontend no longer exposes insecure preset buttons or inline credentials, and the admin Users page can rotate or reset panel account passwords. That shows the database is not just a backend schema; it powers a real management interface.

Appendix G walks through the hardware and the web interface visually.

6. MongoDB Database Design

6.1 Why MongoDB Was the Right Fit

MongoDB was a good choice for this project for several reasons.

First, the project stores different types of data with different stability and structure. User documents are stable and predictable. Credentials are slightly more varied. Schedules include embedded arrays. Raw events are heterogeneous. Diagnostic logs can contain optional fields. MongoDB handles this mixture naturally.

Second, MongoDB makes it easy to model data according to access patterns. The official MongoDB modeling guidance emphasizes that data accessed together should be stored together and that embedding or referencing should be chosen according to application behavior. That matches this project well.

Third, the project was evolving quickly while the prototype was being built. A flexible document model helped me iterate without constantly rewriting a rigid schema and migration logic.

6.2 Database Name and Collections

The database is called osdp_access. The main collections are:

1. users
2. panel_users
3. credentials
4. events
5. access_log
6. readers
7. schedules
8. system_logs

The model layer is implemented in models.py.

6.3 Collection Responsibilities

users

Stores each person or operator in the system. Important fields include username, full name, role, active flag, allowed readers, schedule name, and creation timestamp.

panel_users

Stores the web-panel login accounts separately from access-control identities. This collection currently seeds two fixed operator accounts: admin with role admin, and demo with role viewer. Keeping this collection separate avoids mixing UI operators with cardholders, PIN users, and access schedules. The application now lets an admin either change one of these panel passwords directly or reset a seeded panel account back to its repository default without recreating the document.

credentials

Stores both card and PIN credentials. A credential references a user by user_id. This collection uses references rather than embedding because users can have multiple credentials with independent lifecycles.

schedules

Stores named schedules such as 24/7 and Weekdays 8-18. The schedule periods are embedded because they are small, bounded, and always accessed together.

readers

Stores the latest state of each reader. This includes online/offline information, secure-channel state, tamper, power, and identification data.

events

Stores raw or normalized hardware events. This collection is intentionally flexible because different event types have different fields.

access_log

Stores the final result of access-control decisions and their reasons. This is the key audit collection for understanding why access was granted or denied.

system_logs

Stores internal diagnostics. This is particularly useful while debugging the prototype, deploying on Raspberry Pi, or tracking serial communication issues.

6.4 Sample Documents

User Document

```
{
  "_id": {"$oid": "665f0000000000000000001"},
  "username": "martin",
  "full_name": "Martin Velichkovski",
  "role": "admin",
  "active": true,
  "allowed_readers": [0],
  "schedule": "24/7",
  "created": {"$date": "2026-05-09T10:00:00Z"}
}
```

Credential Document

```
{
  "_id": {"$oid": "665f0000000000000000010"},
  "user_id": {"$oid": "665f0000000000000000001"},
  "type": "card",
  "card_hex": "04A1B2C3D4",
```

```
  "card_dec": "19938448340",
  "bits": 34,
  "format": 0,
  "reader": 0,
  "enrolled": {"$date": "2026-05-09T10:03:00Z"},
  "active": true
}
```

Schedule Document

```
{
  "_id": {"$oid": "665f0000000000000000020"},
  "name": "Weekdays 8-18",
  "periods": [
    {"days": [0, 1, 2, 3, 4], "start": "08:00", "end": "18:00"}
  ]
}
```

Reader Document

```
{
  "_id": {"$oid": "665f0000000000000000030"},
  "index": 0,
  "addr": 0,
  "state": "ONLINE",
  "sc": 0,
  "tamper": 0,
  "power": 0,
  "vendor": "E41E0A",
  "model": 1,
  "serial": "21AA0145",
  "firmware": "2.83.0",
  "last_seen": {"$date": "2026-05-09T10:15:00Z"}
}
```

Access Log Document

```
{
  "_id": {"$oid": "665f0000000000000000040"},
  "ts": {"$date": "2026-05-09T10:20:00Z"},
  "card_hex": "04A1B2C3D4",
  "user_id": {"$oid": "665f0000000000000000001"},
  "username": "martin",
  "granted": true,
  "reader": 0,
  "reason": "credential matched and schedule allowed"
}
```

6.5 Index Strategy

The index strategy in models.py is simple but important.

Collection	Index	Purpose
users	username unique	Prevent duplicates and support quick lookup
panel_users	username unique	Unique web-panel login names
credentials	user_id	List credentials for one user
credentials	card_hex	Fast card lookup during access
events	ts descending	Efficient recent-event queries
access_log	ts descending	Efficient recent audit queries
readers	index unique	Exactly one reader snapshot per index
schedules	name unique	Schedule reuse by name
system_logs	ts descending	Efficient recent diagnostic queries

6.6 Embedding Versus Referencing

The project uses both patterns.

Embedding is used for:

1. allowed_readers inside a user document,
2. periods inside a schedule document.

Referencing is used for:

1. credentials.user_id -> users._id

The rationale is straightforward:

1. schedules and reader lists are small and naturally belong with the parent record,
2. credentials are independent objects that can be created, revoked, queried, and filtered separately.

6.7 Business Rules Enforced by the Application

The application currently enforces the following important rules:

1. usernames must be unique,
2. panel usernames must be unique,
3. schedule names must be unique,
4. readers are uniquely identified by numeric index,
5. disabled users cannot gain access,
6. the demo web account is read-only,
7. admin panel writes require the admin web role,
8. users with empty allowed_readers are allowed on all readers,
9. users with a non-empty list are restricted to those readers,
10. schedules gate access according to current time,

11. raw event history is append oriented,
12. access results are append oriented,
13. reader state is maintained by upsert rather than append-only logging.

6.8 Mongo Shell Representation of the Same Model

Although the main application uses Python and PyMongo, the same database model is also expressed directly in native MongoDB shell syntax through scripts/osdp_access_mongo.js. From a database-course perspective, this shows that the schema and business-oriented data operations are not tied to one application framework. The core model can also be presented directly through mongosh methods such as createCollection(), createIndex(), insertOne(), findOne(), updateOne(), deleteOne(), and deleteMany().

This shell implementation follows the same collection structure described above: users, panel_users, credentials, events, access_log, readers, schedules, and system_logs. It also seeds the same default schedules, adds the same fixed admin and demo panel accounts, includes reset helpers for restoring seeded panel passwords, applies the same lookup rules for cards and PINs, and keeps the same distinction between current reader state and append-only operational logs. In other words, the Mongo shell version is not a separate design; it is a direct representation of the same database model in Mongo-native syntax.

For demonstration purposes, the project also includes scripts/osdp_access_mongo_demo.js, which runs the helper library against a separate database named osdp_access_demo. That allows the project to demonstrate schema creation, CRUD operations, schedule checks, event logging, access evaluation, and audit-log generation without modifying the live application database. The full explanation and the full captured run output are included directly in Appendix E and Appendix F of this report, based on docs/MONGO_SCRIPT_WALKTHROUGH.md and docs/MONGO_SCRIPT_DEMO_OUTPUT.txt.

This connects the conceptual model, the application implementation, and the native MongoDB shell operations in one place. A reader can therefore understand the project at three levels:

1. as a document-model design,
2. as an application-backed MongoDB implementation,
3. as a Mongo shell demonstration of the same schema and logic.

7. Implementation Walkthrough with Commands and Code

7.1 Local Development Commands and What They Mean

Create a virtual environment

```
python -m venv .venv
```

Explanation:

1. python starts the Python interpreter.
2. -m venv tells Python to run the standard-library virtual environment module.
3. .venv is the directory where the isolated environment is created.

Install backend dependencies

```
.venv/bin/pip install -r requirements.txt
```

Explanation:

1. .venv/bin/pip uses the package installer from the project-specific environment.
2. install tells pip to install packages.
3. -r requirements.txt reads the package list from the requirements file.

Build the React frontend

```
cd frontend
npm ci
npm run build
```

Explanation:

1. cd frontend enters the frontend project directory.
2. npm ci installs exactly the versions recorded in package-lock.json.
3. npm run build executes the Vite production build and outputs the files into static/dist.

Run the backend

```
cd ..
.venv/bin/python app.py
```

Explanation:

1. cd .. returns to the repository root.
2. .venv/bin/python uses the Python interpreter from the virtual environment.
3. app.py starts the Flask and Socket.IO backend.

After the backend starts, the web panel now requires login. The two seeded accounts are:

1. admin / osdp for full control,
2. demo / db2 for a read-only viewer session intended for demonstrations.

For better security, the frontend does not reveal those defaults on the login screen anymore. If the passwords were rotated earlier, an admin can restore them either from the Users page or through the Mongo shell helpers before a demonstration.

For temporary off-site access, the repository also includes share-ngrok.ps1, which starts ngrok http 5000 and prints a public HTTPS URL for the local panel. That makes it possible to demonstrate the dashboard and live logs safely through the demo account without exposing write operations.

7.2 Raspberry Pi Deployment Command and Line-by-Line Meaning

The most important deployment command is the Docker run used for MongoDB.

```
docker run -d \
  --name osdp-access-mongo \
  --restart unless-stopped \
  -p 127.0.0.1:27017:27017 \
```

```
  -v /home/admin/osdp-access-mongo:/data/db \
  mongo:4.4.18
```

Explanation:

1. docker run creates and starts a new container.
2. -d starts it in detached background mode.
3. --name osdp-access-mongo gives the container a stable name.
4. --restart unless-stopped makes it restart automatically after reboot unless explicitly stopped.
5. -p 127.0.0.1:27017:27017 binds MongoDB only to localhost on the Pi for safer local-only access.
6. -v /home/admin/osdp-access-mongo:/data/db stores MongoDB data persistently on the Pi filesystem.
7. mongo:4.4.18 chooses the MongoDB image tag.

That version choice matters. During deployment, newer MongoDB images failed on the target Pi because of ARM CPU feature requirements, so the setup was pinned to a compatible version.

7.3 Backup and Restore Commands and What They Mean

Backup

```
python backup_mongo.py --mongo-uri mongodb://localhost:27017 --db osdp_access --output-dir back
ups
```

Explanation:

1. python backup_mongo.py runs the backup tool.
2. --mongo-uri mongodb://localhost:27017 tells the script where MongoDB is running.
3. --db osdp_access selects the database to export.
4. --output-dir backups selects the root folder where the timestamped backup directory will be created.

Restore

```
python restore_mongo.py backups/mongodb_osdp_access_YYYYMMDD_HHMMSS --mongo-uri mongodb://local
host:27017
```

Explanation:

1. python restore_mongo.py runs the restore tool.
2. backups/... is the specific backup directory that contains manifest.json and collection dumps.
3. --mongo-uri ... points to the target MongoDB instance.

7.4 Annotated Code Example: MongoDB Initialization

The following excerpt from models.py is the entry point into the database layer.

```
def get_db(uri="mongodb://localhost:27017"):
    client = MongoClient(uri, serverSelectionTimeoutMS=3000)
    db = client[DB_NAME]
    _ensure_indexes(db)
    return db
```

Line-by-line explanation:

1. def get_db(...): defines a helper that returns the database handle used everywhere else.
2. uri="mongodb://localhost:27017" provides a default local MongoDB address.
3. MongoClient(uri, serverSelectionTimeoutMS=3000) creates the MongoDB client and fails quickly if the database is not reachable.
4. db = client[DB_NAME] selects the osdp_access database.
5. _ensure_indexes(db) creates indexes and default schedules automatically.
6. return db gives the caller a ready-to-use database handle.

The initialization logic continues with _ensure_indexes():

```
def _ensure_indexes(db):
    db.users.create_index("username", unique=True)
  db.panel_users.create_index("username", unique=True)
    db.credentials.create_index("user_id")
    db.credentials.create_index("card_hex")
    db.events.create_index([("ts", DESCENDING)])
    db.access_log.create_index([("ts", DESCENDING)])
    db.readers.create_index("index", unique=True)
    db.schedules.create_index("name", unique=True)
    db.system_logs.create_index([("ts", DESCENDING)])
```

Line-by-line explanation:

1. db.users.create_index("username", unique=True) prevents duplicate usernames.
2. db.panel_users.create_index("username", unique=True) prevents duplicate login names for the web panel.
3. db.credentials.create_index("user_id") accelerates queries for all credentials of one user.
4. db.credentials.create_index("card_hex") makes card lookups fast during access decisions.
5. db.events.create_index([("ts", DESCENDING)]) makes recent-event pages efficient.
6. db.access_log.create_index([("ts", DESCENDING)]) makes audit-log pages efficient.
7. db.readers.create_index("index", unique=True) ensures one state document per reader.
8. db.schedules.create_index("name", unique=True) prevents duplicate schedule names.
9. db.system_logs.create_index([("ts", DESCENDING)]) supports recent diagnostic log queries.

7.5 Annotated Code Example: Serial Bridge Connection

The bridge connection logic in bridge.py is one of the most important pieces of the project.

```
def connect(self, port=None, retries=3):
    if self.ser and self.ser.is_open and self.connected:
        return True
    self._cleanup()
    port = port or self.find_port()
    if not port:
        log.warning("No Blue Pill port found")
        return False
```

Line-by-line explanation:

1. connect(..., retries=3) attempts to open the bridge with automatic retries.

2. The first if avoids reconnecting if the bridge is already usable.
3. self._cleanup() clears stale serial state before a new attempt.
4. port = port or self.find_port() uses the explicitly provided port or auto-detects the Blue Pill by USB VID and PID.
5. If no port is found, the method logs a warning and returns False.

This is important because real USB hardware is unreliable in the field: ports disappear, reconnect, or enumerate differently after reboot.

7.6 Annotated Code Example: Parsing Bridge Events

The bridge converts raw lines from the STM32 into structured Python dictionaries. A representative example is the card event parser.

```
m = re.match(r"!CARD (\d+) ([0-9A-Fa-f]+) (\d+) (\d+)", line)
if m:
    return {
        "type": "card",
        "reader": int(m.group(1)),
        "hex": m.group(2),
        "bits": int(m.group(3)),
        "format": int(m.group(4)),
        "ts": ts,
        "raw": line,
    }
```

Line-by-line explanation:

1. re.match(...) looks for a line beginning with !CARD followed by reader number, card hex value, bit length, and format.
2. if m: means the parser only continues if the pattern matched.
3. "type": "card" normalizes the event into an internal event type.
4. "reader": int(...) records which reader generated the card event.
5. "hex": ... stores the raw credential value used for lookup or enrollment.
6. "bits": ... preserves bit length metadata.
7. "format": ... preserves format metadata.
8. "ts": ts records a backend-side timestamp.
9. "raw": line preserves the original line for diagnostics.

This is a good example of why MongoDB fits the project: different event types naturally carry different fields, and a flexible document model stores them without a complicated rigid schema.

7.7 Annotated Code Example: Card Access Decision

The access-policy path in app.py is central to the controller logic.

```
cred = find_credential_by_card(db, ev["hex"])
if cred:
    user = get_user(db, str(cred["user_id"]))
    name = user["username"] if user else "unknown"
    allowed, reason = _check_access_policy(user, reader) if user else (False, "user not found")
```

```
    if allowed:
        log_access(db, card_hex=ev["hex"], user_id=str(cred["user_id"]), username=name,
                   granted=True, reader=reader, reason=reason)
        bridge.relay(reader, "T1500")
        bridge.grant_feedback(reader)
```

Line-by-line explanation:

1. find_credential_by_card(...) looks up the presented card in MongoDB.
2. if cred: continues only if the card exists and is active.
3. get_user(...) loads the user referenced by the credential.
4. name = ... safely produces a display name even if the user is missing.
5. _check_access_policy(...) applies business rules such as active state, allowed readers, and schedule.
6. log_access(...) writes the final audit record into access_log.
7. bridge.relay(...) activates the relay for a timed pulse.
8. bridge.grant_feedback(...) turns that decision into LED and buzzer behavior at the reader.

This short code path demonstrates the whole idea of the project: field input becomes database lookup, policy evaluation, audit logging, and real physical output.

7.8 Annotated Code Example: Firmware Flashing

The STM32 flasher in flasher.py is a good example of tooling around the hardware platform.

```
resp = _send_cmd(ser, "ERASE", timeout=30)
if resp != "OK":
    raise RuntimeError(f"Erase failed: {resp}")

for i in range(total_chunks):
    offset = i * CHUNK_SIZE
    chunk = fw_data[offset : offset + CHUNK_SIZE]
    offset_hex = f"{offset:05X}"
    data_hex = chunk.hex().upper()
    cmd = f"W{offset_hex} {data_hex}"
    resp = _send_cmd(ser, cmd, timeout=5)
```

Line-by-line explanation:

1. _send_cmd(ser, "ERASE", timeout=30) instructs the bootloader to erase the application flash region.
2. if resp != "OK": checks that the bootloader acknowledged the command.
3. for i in range(total_chunks): starts chunked upload of the firmware image.
4. offset = i * CHUNK_SIZE computes the byte offset for each block.
5. chunk = ... slices the firmware bytes for the current block.
6. offset_hex = ... converts the offset into the format expected by the bootloader.
7. data_hex = chunk.hex().upper() converts binary bytes into ASCII hex.
8. cmd = f"W{offset_hex} {data_hex}" builds the bootloader write command.
9. _send_cmd(...) sends the write operation and waits for confirmation.

This tool makes the hardware platform maintainable and is an important part of the engineering completeness of the project.

7.9 Annotated Raspberry Pi Setup Script

The Raspberry Pi deployment script in scripts/setup_raspberry_pi.sh is worth understanding because it turns the project into a reproducible platform.

```
ROOT="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
MONGO_URI="${MONGO_URI:-mongodb://localhost:27017}"
MONGO_IMAGE="${MONGO_IMAGE:-mongo:4.4.18}"
```

Line-by-line explanation:

1. ROOT=... computes the repository root directory automatically.
2. MONGO_URI=... defines the MongoDB connection string, with a default value if none is provided.
3. MONGO_IMAGE=... allows the script to override the Docker image while defaulting to a Pi-compatible tag.

Later in the script, the database container is created, the Python environment is prepared, the frontend is built, and the app is registered as a systemd service. That makes the deployment path self-documenting and repeatable.

8. Representative MongoDB Queries and Reports

8.1 Find All Active Users

```
db.users.find(
  { active: true },
  { username: 1, full_name: 1, role: 1, schedule: 1 }
).sort({ username: 1 })
```

This query retrieves currently active users, sorted alphabetically.

8.2 Find Credentials for a Specific User

```
db.credentials.find({
  user_id: ObjectId("665f0000000000000000001"),
  active: true
}).sort({ enrolled: -1 })
```

This query is useful in the UI when reviewing or revoking user credentials.

8.3 Retrieve the Latest Reader State

```
db.readers.find().sort({ index: 1 })
```

This query feeds the reader dashboard with current state snapshots instead of parsing the whole event history.

8.4 Count Events by Type

```
db.events.aggregate([
  { $group: { _id: "$type", count: { $sum: 1 } } },
  { $sort: { count: -1 } }
])
```

This query is useful when analyzing whether the system is mostly seeing card traffic, keypad traffic, status updates, or configuration events.

8.5 Count Denied Access Reasons

```
db.access_log.aggregate([
  { $match: { granted: false } },
  { $group: { _id: "$reason", count: { $sum: 1 } } },
  { $sort: { count: -1 } }
])
```

This query is valuable operationally because it explains why the system is denying access: disabled users, outside schedule, reader not allowed, unknown card, or unknown PIN.

8.6 Count Access Attempts Per Reader

```
db.access_log.aggregate([
  { $group: { _id: "$reader", attempts: { $sum: 1 } } },
  { $sort: { attempts: -1 } }
])
```

This query helps analyze which reader is most active on the test bench or in future multi-reader deployments.

9. Deployment, Testing, and Development Story

9.1 Local Validation

The project was validated locally through:

1. backend startup tests,
2. login and role enforcement tests for admin and demo,
3. reader connection tests,
4. interactive OSDP workflow tests,
5. MongoDB backup generation,
6. MongoDB restore validation,
7. frontend build validation,
8. temporary public exposure through an ngrok tunnel for read-only review.

9.2 Raspberry Pi Deployment

The project was also deployed on a Raspberry Pi. This was important because it moved the project from a desktop development setup into a more controller-like environment.

During deployment, two real engineering issues appeared:

1. newer MongoDB container images were incompatible with the Pi CPU,
2. Flask-SocketIO required explicit allow_unsafe_werkzeug=True for the current deployment approach.

Those issues were fixed, documented, and folded back into the repository. That is a strong example of why real-world database work is inseparable from deployment and operations.

9.3 Current Working Result

At the time of writing:

1. the system runs on the Pi,
2. MongoDB is running in Docker,
3. the frontend is served successfully,
4. the backend responds on HTTP,
5. the bridge connects to the STM32,
6. the reader is visible and reporting status,
7. data backup and restore work.

9.4 Personal Engineering Reflection

This project became more than a database assignment because the database work was learned and applied in parallel with hardware assembly and embedded debugging. That made the process more demanding, but it also made the results easier to evaluate in a practical way. Instead of studying NoSQL concepts only in theory, I applied them while building a physical controller, wiring boards, debugging reader communication, and developing a tool that is useful in my day-to-day engineering context.

That is also why the project reflects real engineering constraints. It was built under time pressure, with physical hardware, firmware iteration, Linux deployment issues, database modeling decisions, and the need to keep all parts of the system working together.

10. Conclusion

This project demonstrates a practical MongoDB application in the domain of access control and embedded systems. The OSDP Access Controller PoC is not only a database model; it is a complete hardware-software platform that uses MongoDB as its operational memory, audit trail, and administrative data store.

The project also shows why MongoDB was a good choice. The system needed to store stable administrative documents, append-heavy event logs, flexible device telemetry, and diagnostic records. MongoDB handles this combination naturally. The final schema is easy to extend, easy to query, and aligned with the access patterns of the application.

From an educational perspective, the project demonstrates what a Databases II project should aim for: using a NoSQL database in a meaningful context, applying design ideas such as embedding versus referencing and index planning, implementing queries and data operations in code, and connecting the database to a

real application stack. From an engineering perspective, it also produced a useful controller test platform for reader development.

Future work could include formal MongoDB schema validation, richer aggregation dashboards, multi-reader topologies, long-term log archival, and a more production-grade WSGI or ASGI deployment path. Even in its current form, however, the project already functions as both an academic NoSQL project and a practical engineering tool.

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Appendix A: Important Files in the Project

File	Role
app.py	Main backend entrypoint

File	Role
bridge.py	Serial bridge client and event parser
models.py	MongoDB model helpers
flasher.py	STM32 bridge firmware updater
osdp-controller/src/main.cpp	Main STM32 bridge firmware entrypoint
osdp-controller/src/osdp_cp.cpp	OSDP control-panel implementation on the MCU
bootloader/src/main.cpp	STM32 USB bootloader entrypoint
backup_mongo.py	Database export tool
restore_mongo.py	Database import tool
run.sh	Linux build and run helper
scripts/setup_raspberry_pi.sh	Raspberry Pi installer
scripts/osdp_access_mongo.js	Standalone mongosh helper library for the full database model
scripts/osdp_access_mongo_demo.js	End-to-end mongosh demo that exercises CRUD and logging features
frontend/src/App.jsx	Top-level React application shell
docs/MONGO_SCRIPT_WALKTHROUGH.md	Mongo-only walkthrough with real demo output

Appendix B: Notes for Final Submission

Before submitting the final Word or PDF version, the following can be customized if needed:

1. team members and student IDs,
2. exact mentor line,
3. insertion of the final prototype photo if desired,
4. conversion to PDF if Moodle prefers PDF,
5. addition of any lecturer-specific formatting requirements.

Appendix C: Mongo-Only Implementation and Demo

Why This Appendix Was Added

For class presentation purposes, it is useful to show the database not only through the Python backend, but also directly through native MongoDB shell syntax. To support that, the project now includes a standalone mongosh implementation of the database layer and a separate demo script that exercises the main database operations without depending on Flask or the React frontend.

The relevant files are:

1. scripts/osdp_access_mongo.js
2. scripts/osdp_access_mongo_demo.js
3. docs/MONGO_SCRIPT_WALKTHROUGH.md, reproduced in Appendix E
4. docs/MONGO_SCRIPT_DEMO_OUTPUT.txt, reproduced in Appendix F

What The Mongo Shell Library Demonstrates

The shell library demonstrates the same database model as the application code, but directly in MongoDB shell syntax. It covers:

1. collection creation and index setup,
2. default schedule seeding,
3. user CRUD,
4. credential CRUD for cards and PINs,
5. schedule CRUD,
6. reader state upserts and reads,
7. raw event logging and queries,
8. access log logging and queries,
9. system log logging and queries,
10. access evaluation helpers for card and PIN workflows,
11. reset and summary helpers.

One important design choice is that the library can work against a different database name through:

```
const demo = createOsdpAccessApi("osdp_access_demo")
```

This makes it possible to run demonstrations safely without modifying the real application database.

How The Demo Was Run

The clean execution command used for the demonstration is:

```
mongosh mongodb://localhost:27017 --quiet --file scripts/osdp_access_mongo.js --file scripts/os
dp_access_mongo_demo.js
```

The command was executed against a dedicated demo database called osdp_access_demo, and the resulting shell output is captured both in docs/MONGO_SCRIPT_DEMO_OUTPUT.txt and directly in Appendix F of this report.

Selected Real Output

The following excerpt from the real run shows that initialization created the expected collections and seeded the default schedules:

```
=== 1. Reset and initialize demo database ===
-- resetDatabase({ dropDatabase: true })
{
  "db": "osdp_access_demo",
  "collections": [
    "access_log",
```

```
    "credentials",
    "events",
    "readers",
    "schedules",
    "system_logs",
    "users"
  ],
  "counts": {
    "users": 0,
    "credentials": 0,
    "events": 0,
    "access_log": 0,
    "readers": 0,
    "schedules": 2,
    "system_logs": 0
  }
}
```

The next excerpt shows schedule-aware evaluation and full access workflow behavior:

```
=== 9. Access evaluation helpers ===
-- evaluateUserAccess(martin, reader 0, mondayMorning)
{
  "granted": true,
  "reason": "allowed"
}
-- evaluateUserAccess(visitor, reader 0, mondayMorning)
{
  "granted": false,
  "reason": "reader not allowed"
}
-- evaluateUserAccess(visitor, reader 1, saturdayNight)
{
  "granted": false,
  "reason": "outside schedule"
}

=== 10. Access workflows and access log ===
-- accessByCard(martin card)
{
  "granted": true,
  "reason": "allowed"
}
-- accessByCard(unknown card)
{
  "granted": false,
  "reason": "unknown card"
}
```

These excerpts show the database logic directly in Mongo shell terms, which is often one of the main points of interest in a database course setting.

Appendix D: MongoDB Shell Operator Map For Helper Functions

This appendix maps each helper function from scripts/osdp_access_mongo.js to the MongoDB shell operators or collection methods it demonstrates.

Core and Setup Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
createOsdpAccessApi	db.getSiblingDB()	Select a target database and build the helper API around it
init	createCollection(), createIndex(), updateOne() with $setOnInsert, countDocuments(), getCollectionNames()	Create schema, indexes, defaults, and summary information
help	none, shell print() only	Print the helper API usage summary
summary	getCollectionNames(), countDocuments()	Return database and collection counts
resetDatabase	dropDatabase(), deleteMany()	Reset the demo or application database
collection	getCollection()	Expose direct collection access for advanced shell work
ensureCollections	getCollectionNames(), createCollection()	Create missing collections
ensureIndexes	createIndex()	Create uniqueness and lookup indexes
seedSchedules	updateOne() with $setOnInsert and upsert: true	Seed default schedule documents
seedPanelUsers	findOne(), insertOne(), updateOne() with $set	Seed or repair fixed panel login accounts

Panel User Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
listPanelUsers	find(), sort(), toArray()	Read panel login accounts in username order
getPanelUserByUsername	findOne()	Read one panel login account by unique username
resetPanelUserPassword	updateOne() with $set	Restore one seeded panel login account to its default password hash
resetAllPanelUserPasswords	repeated updateOne() with $set	Restore every seeded panel login account to its default password hash

User Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
createUser	insertOne()	Insert a user document
listUsers	find(), sort(), toArray()	Read users in username order
getUserById	findOne()	Read a user by _id
getUserByUsername	findOne()	Read a user by unique username
updateUser	updateOne() with $set	Update selected user fields
deactivateUser	updateOne() with $set	Soft-delete a user by setting active: false
deleteUser	deleteMany(), deleteOne()	Remove a user and optionally cascade credentials

Credential Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
enrollCard	insertOne()	Insert a card credential
enrollPin	insertOne()	Insert a PIN credential
listCredentials	find(), sort(), toArray()	Read credential documents
getCredentialById	findOne()	Read one credential by _id
updateCredential	updateOne() with $set	Update card, PIN, or owner fields
revokeCredential	updateOne() with $set	Soft-revoke a credential
deleteCredential	deleteOne()	Hard-delete a credential
findCredentialByCard	findOne()	Find an active card credential
findCredentialByPin	findOne()	Find an active PIN credential

Schedule and Policy Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
listSchedules	find(), sort(), toArray()	Read all schedules
getSchedule	findOne()	Read a named schedule
createSchedule	insertOne()	Insert a schedule document

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
updateSchedule	updateOne() with $set	Modify schedule periods or metadata
deleteSchedule	deleteOne()	Remove a schedule
checkSchedule	findOne()	Evaluate time-based access against stored schedule data
checkReaderAccess	none, in-memory rule evaluation	Check allowed_readers against a reader index
evaluateUserAccess	none directly; uses checkReaderAccess() and checkSchedule()	Return the final access decision and reason

Event, Access Log, and System Log Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
logEvent	insertOne()	Insert a raw or normalized event document
getEvents	find(), sort(), limit(), toArray()	Query recent event history
deleteEvents	deleteMany()	Remove selected event rows
logAccess	insertOne()	Insert an audit log entry
getAccessLog	find(), sort(), limit(), toArray()	Read recent access decisions
deleteAccessLog	deleteMany()	Remove selected access log entries
logSystem	insertOne()	Insert a system or diagnostic log
getSystemLogs	find(), sort(), limit(), toArray()	Query recent system logs
deleteSystemLogs	deleteMany()	Remove selected diagnostic logs

Reader Snapshot Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
upsertReader	updateOne() with $set and upsert: true	Maintain one current-state document per reader
getReader	findOne()	Read a single reader snapshot
listReaders	find(), sort(), toArray()	Read all reader snapshots
deleteReader	deleteOne()	Remove a reader snapshot

Full Access Workflow Helpers

Helper Function	MongoDB Shell Operators / Methods Demonstrated	Purpose
accessByCard	findOne() via credential and user lookup, insertOne() via logAccess()	Execute full card-based access evaluation and audit logging
accessByPin	findOne() via credential and user lookup, insertOne() via logAccess()	Execute full PIN-based access evaluation and audit logging

Why This Table Is Useful For Assessment

This operator map makes the shell library easier to assess in a classroom setting because it shows exactly which MongoDB methods are being demonstrated by each helper. In other words, the project is not only using MongoDB through an application framework; it can also be explained directly in terms of native MongoDB shell operations.

Appendix E: Full Mongo Shell Walkthrough

This appendix reproduces the standalone walkthrough from docs/MONGO_SCRIPT_WALKTHROUGH.md so the report contains the narrative explanation together with the main body, instead of requiring a separate file.

Why This Matters

For database-course purposes, this appendix presents a database implementation expressed directly in native MongoDB shell syntax. Instead of describing MongoDB concepts only at the application level, this walkthrough shows the project as a pure mongosh solution.

The two files behind this walkthrough are:

1. scripts/osdp_access_mongo.js - the reusable MongoDB helper library
2. scripts/osdp_access_mongo_demo.js - the end-to-end demonstration script

The library is modeled directly in Mongo shell operations such as:

1. createCollection() for explicit collection creation
2. createIndex() for uniqueness and read performance
3. insertOne() for creates
4. find() and findOne() for reads
5. updateOne() for updates and upserts
6. deleteOne() and deleteMany() for deletes
7. countDocuments() for database summaries

The goal is to show that the whole database side of the project can be understood and demonstrated without relying on the Python backend.

What The Library Implements

The helper library provides all core database operations used by the project:

1. collection creation and index setup
2. default schedule seeding
3. user CRUD
4. credential CRUD for cards and PINs
5. schedule CRUD
6. reader state upserts and reads
7. raw event logging and queries
8. access log logging and queries
9. system log logging and queries
10. access evaluation helpers for card and PIN workflows
11. reset and summary helpers

One important addition is that the library now supports a factory function:

```
const demo = createOsdpAccessApi("osdp_access_demo")
```

That allows the same logic to be run safely against a separate demo database, which is exactly what the demonstration script uses.

How To Run It

Load the library only

```
mongosh mongodb://localhost:27017/osdp_access
load("scripts/osdp_access_mongo.js")
osdpAccess.init()
osdpAccess.help()
```

Run the full demo

```
mongosh mongodb://localhost:27017 --quiet --file scripts/osdp_access_mongo.js --file scripts/os
dp_access_mongo_demo.js
```

The full raw transcript from the real run is stored in docs/MONGO_SCRIPT_DEMO_OUTPUT.txt and is also reproduced in Appendix F.

Script Design In Words

1. Initialization Layer

The library starts by defining the target collection set and the two default schedules used by the access-control logic. The init() function creates collections if needed, creates indexes, and seeds schedules. This mirrors the role of _ensure_indexes() from the Python backend, but it is now expressed directly in mongosh.

2. User Layer

The user API models a person or operator with username, full_name, role, active, allowed_readers, schedule, and created. The design uses a soft-delete style through deactivateUser() and also exposes deleteUser() for hard deletion.

3. Credential Layer

Card and PIN credentials are stored in the credentials collection and reference users through user_id. The library normalizes card and PIN values to uppercase and derives card_dec from card_hex so the data matches the backend schema.

4. Schedule Layer

Schedules are stored as named documents with embedded periods. The helper functions demonstrate create, read, update, delete, and also direct time evaluation through checkSchedule().

5. Reader Layer

The readers collection is not append-only. It stores the latest known snapshot for each reader. That is why the library uses upsertReader() instead of inserting a new document every time state changes.

6. Logging Layer

The database keeps three different categories of operational data separate:

1. events for raw or normalized MCU and bridge events
2. access_log for final access decisions
3. system_logs for diagnostics and troubleshooting messages

That separation is visible in the library design and also in the demo output.

7. Access Evaluation Layer

The most interesting part from an access-control perspective is the evaluation logic. The script shows that an access decision is not just a credential lookup. It also depends on:

1. whether the user exists
2. whether the user is active
3. whether the user is allowed on the specific reader
4. whether the assigned schedule currently allows access

The accessByCard() and accessByPin() helpers execute the full lookup-and-log flow and then write the result into access_log.

Demo Walkthrough With Real Output

The following sections describe what each demo step does and show the actual output produced by the live mongosh run.

Step 1: Reset and Initialize Demo Database

This step proves that the demo database starts clean and that initialization creates the expected collections and default schedules.

```
=== 1. Reset and initialize demo database ===
-- resetDatabase({ dropDatabase: true })
{
  "db": "osdp_access_demo",
  "collections": [
    "access_log",
    "credentials",
    "events",
    "readers",
    "schedules",
    "system_logs",
    "users"
  ],
  "counts": {
    "users": 0,
    "credentials": 0,
    "events": 0,
    "access_log": 0,
    "readers": 0,
    "schedules": 2,
    "system_logs": 0
  }
}
```

Step 2: Create Users

This step inserts three demo users and then reads them back.

```
=== 2. Create users ===
-- users after create
[
  {
    "username": "martin",
    "full_name": "Martin Velichkovski",
    "role": "admin",
    "active": true,
    "allowed_readers": [
      0
    ],
    "schedule": "24/7"
  },
  {
    "username": "temp-delete",
    "full_name": "Temporary Demo User",
    "role": "user",
    "active": true,
    "allowed_readers": [],
    "schedule": "24/7"
  },
```

```
  {
    "username": "visitor",
    "full_name": "Weekend Visitor",
    "role": "user",
    "active": true,
    "allowed_readers": [
      1
    ],
    "schedule": "Weekdays 8-18"
  }
]
```

Step 3: Read and Update Users

This step demonstrates both read operations and a soft-delete style update.

```
=== 3. Read and update users ===
-- getUserByUsername("martin")
{
  "username": "martin",
  "full_name": "Martin Velichkovski",
  "role": "admin",
  "active": true
}
-- updateUser(martin)
{
  "acknowledged": true,
  "matchedCount": 1,
  "modifiedCount": 1
}
-- getUserById(martinId)
{
  "username": "martin",
  "full_name": "Martin V.",
  "allowed_readers": [
    0,
    2
  ]
}
-- deactivateUser(tempUserId)
{
  "acknowledged": true,
  "matchedCount": 1,
  "modifiedCount": 1
}
```

Step 4: Credential CRUD

This step creates card and PIN credentials, queries them, updates one, revokes another, and then deletes it.

```
=== 4. Create, read, update, revoke, and delete credentials ===
-- findCredentialByCard("04A1B2C3D4")
{
  "type": "card",
  "card_hex": "04A1B2C3D4",
```

```
  "card_dec": "19892716500",
  "bits": 34,
  "format": 0,
  "reader": 0,
  "active": true
}
-- updateCredential(martinCardId)
{
  "acknowledged": true,
  "matchedCount": 1,
  "modifiedCount": 1
}
-- getCredentialById(martinCardId)
{
  "type": "card",
  "card_hex": "04A1B2C3D4",
  "bits": 37,
  "reader": 2,
  "active": true
}
-- revokeCredential(tempCredentialId)
{
  "acknowledged": true,
  "matchedCount": 1,
  "modifiedCount": 1
}
-- deleteCredential(tempCredentialId)
{
  "acknowledged": true,
  "deletedCount": 1
}
```

Step 5: Schedule CRUD and Time Evaluation

This step shows schedule creation, schedule update, direct schedule evaluation, and deletion.

```
=== 5. Create, read, update, delete schedules ===
-- checkSchedule("Weekdays 8-18", mondayMorning)
{
  "schedule": "Weekdays 8-18",
  "allowed": true
}
-- checkSchedule("Weekdays 8-18", saturdayNight)
{
  "schedule": "Weekdays 8-18",
  "allowed": false
}
-- deleteSchedule(deleteMeScheduleId)
{
  "acknowledged": true,
  "deletedCount": 1
}
```

Step 6: Reader Snapshot Management

This step demonstrates why readers are stored as current state snapshots rather than append-only log rows.

```
=== 6. Upsert, read, list, and delete readers ===
-- upsertReader(0, ...)
{
  "acknowledged": true,
  "matchedCount": 0,
  "modifiedCount": 0,
  "upsertedCount": 1
}
-- getReader(0)
{
  "index": 0,
  "addr": 0,
  "state": "ONLINE",
  "sc": 1,
  "serial": "21AA0145",
  "firmware": "2.83.0"
}
-- deleteReader(1)
{
  "acknowledged": true,
  "deletedCount": 1
}
```

Step 7: Event Logging and Cleanup

This step logs raw events, reads them back, and deletes a test event.

```
=== 7. Log, query, and delete events ===
-- getEvents({ limit: 10 })
[
  {
    "type": "card",
    "reader": 0,
    "hex": "04A1B2C3D4",
    "raw": "!CARD demo"
  },
  {
    "type": "pd_status",
    "reader": 0,
    "state": "ONLINE",
    "raw": "!PD 0 ONLINE"
  },
  {
    "type": "demo_delete",
    "reader": 9,
    "raw": "!DEMO DELETE"
  }
]
-- deleteEvents({ type: "demo_delete" })
{
```

```
  "acknowledged": true,
  "deletedCount": 1
}
```

Step 8: System Log CRUD

This step shows that diagnostic logs are handled separately from raw device events and access decisions.

```
=== 8. Log, query, and delete system logs ===
-- getSystemLogs({ limit: 10 })
[
  {
    "level": "info",
    "source": "demo-cleanup",
    "message": "This log will be deleted"
  },
  {
    "level": "warn",
    "source": "demo",
    "message": "Reader 1 was removed during cleanup"
  },
  {
    "level": "info",
    "source": "demo",
    "message": "Demo script started"
  }
]
-- deleteSystemLogs({ source: "demo-cleanup" })
{
  "acknowledged": true,
  "deletedCount": 1
}
```

Step 9: Access Evaluation Helpers

This step is important conceptually because it shows that access depends on more than a credential lookup.

```
=== 9. Access evaluation helpers ===
-- evaluateUserAccess(martin, reader 0, mondayMorning)
{
  "granted": true,
  "reason": "allowed"
}
-- evaluateUserAccess(visitor, reader 0, mondayMorning)
{
  "granted": false,
  "reason": "reader not allowed"
}
-- evaluateUserAccess(visitor, reader 1, saturdayNight)
{
  "granted": false,
  "reason": "outside schedule"
}
```

Step 10: Full Access Workflows and Audit Trail

This step runs the complete accessByCard() and accessByPin() workflows and then reads the resulting audit log.

```
=== 10. Access workflows and access log ===
-- accessByCard(martin card)
{
  "granted": true,
  "reason": "allowed"
}
-- accessByCard(unknown card)
{
  "granted": false,
  "reason": "unknown card"
}
-- accessByPin(visitor pin outside schedule)
{
  "granted": false,
  "reason": "outside schedule"
}
-- getAccessLog({ limit: 10 })
[
  {
    "card_hex": "DEADBEEF",
    "granted": false,
    "reader": 99,
    "reason": "manual demo delete"
  },
  {
    "pin_hex": "BEEF",
    "username": "visitor",
    "granted": false,
    "reason": "outside schedule"
  },
  {
    "card_hex": "FFFFFFFF",
    "granted": false,
    "reason": "unknown card"
  },
  {
    "card_hex": "04A1B2C3D4",
    "username": "martin",
    "granted": true,
    "reason": "allowed"
  }
]
```

Step 11: Cleanup and Final Summary

This step removes the temporary user and prints the final database counts.

```
=== 11. Delete temporary user and final summary ===
-- deleteUser(tempUserId)
```

```
{
  "acknowledged": true,
  "deletedCount": 1
}
-- final summary
{
  "db": "osdp_access_demo",
  "counts": {
    "users": 2,
    "credentials": 3,
    "events": 2,
    "access_log": 3,
    "readers": 1,
    "schedules": 3,
    "system_logs": 2
  }
}
```

What This Demonstration Shows

From a class perspective, the important point is that the complete database workflow can now be shown directly in MongoDB shell syntax, without needing Flask, Python, or the frontend.

The demonstration proves that the MongoDB design supports:

1. creation of the access-control schema and indexes
2. real CRUD operations on administrative entities
3. logging of heterogeneous runtime events
4. separation between raw logs, access decisions, and diagnostics
5. schedule-aware and reader-aware access decisions
6. repeatable testing through a dedicated demo database

Files Produced

1. scripts/osdp_access_mongo.js
2. scripts/osdp_access_mongo_demo.js
3. docs/MONGO_SCRIPT_DEMO_OUTPUT.txt

These three files together give you:

1. the reusable Mongo-only implementation
2. the live demonstration script
3. the captured shell output for documentation and submission support

Appendix F: Full Mongo Shell Demo Transcript

This appendix reproduces the full raw output from docs/MONGO_SCRIPT_DEMO_OUTPUT.txt so the report contains the actual terminal transcript that produced the Mongo shell demonstration.

```
Loaded osdpAccess Mongo helpers. Run osdpAccess.help() for usage.

=== 1. Reset and initialize demo database ===
-- resetDatabase({ dropDatabase: true })
{
  "db": "osdp_access_demo",
  "collections": [
    "access_log",
    "credentials",
    "events",
    "readers",
    "schedules",
    "system_logs",
    "users"
  ],
  "counts": {
    "users": 0,
    "credentials": 0,
    "events": 0,
    "access_log": 0,
    "readers": 0,
    "schedules": 2,
    "system_logs": 0
  }
}

=== 2. Create users ===
-- users after create
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "username": "martin",
    "full_name": "Martin Velichkovski",
    "role": "admin",
    "active": true,
    "allowed_readers": [
      0
    ],
    "schedule": "24/7",
    "created": {
      "$date": "2026-05-09T00:45:55.531Z"
    }
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2909"
    },
    "username": "temp-delete",
    "full_name": "Temporary Demo User",
    "role": "user",
    "active": true,
```

```
    "allowed_readers": [],
    "schedule": "24/7",
    "created": {
      "$date": "2026-05-09T00:45:55.533Z"
    }
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2908"
    },
    "username": "visitor",
    "full_name": "Weekend Visitor",
    "role": "user",
    "active": true,
    "allowed_readers": [
      1
    ],
    "schedule": "Weekdays 8-18",
    "created": {
      "$date": "2026-05-09T00:45:55.532Z"
    }
  }
]

=== 3. Read and update users ===
-- getUserByUsername("martin")
{
  "_id": {
    "$oid": "69fe83c33d739553f17c2907"
  },
  "username": "martin",
  "full_name": "Martin Velichkovski",
  "role": "admin",
  "active": true,
  "allowed_readers": [
    0
  ],
  "schedule": "24/7",
  "created": {
    "$date": "2026-05-09T00:45:55.531Z"
  }
}
-- updateUser(martin)
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}
-- getUserById(martinId)
{
  "_id": {
```

```
    "$oid": "69fe83c33d739553f17c2907"
  },
  "username": "martin",
  "full_name": "Martin V.",
  "role": "admin",
  "active": true,
  "allowed_readers": [
    0,
    2
  ],
  "schedule": "24/7",
  "created": {
    "$date": "2026-05-09T00:45:55.531Z"
  }
}
-- deactivateUser(tempUserId)
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}
-- temp user after deactivate
{
  "_id": {
    "$oid": "69fe83c33d739553f17c2909"
  },
  "username": "temp-delete",
  "full_name": "Temporary Demo User",
  "role": "user",
  "active": false,
  "allowed_readers": [],
  "schedule": "24/7",
  "created": {
    "$date": "2026-05-09T00:45:55.533Z"
  }
}

=== 4. Create, read, update, revoke, and delete credentials ===
-- listCredentials()
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c290d"
    },
    "user_id": {
      "$oid": "69fe83c33d739553f17c2909"
    },
    "type": "card",
    "card_hex": "0BADF00D",
    "card_dec": "195948557",
    "bits": 32,
```

```
    "format": 1,
    "reader": 3,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.545Z"
    },
    "active": true
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c290b"
    },
    "user_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "type": "pin",
    "pin_hex": "1A2B",
    "reader": 0,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.544Z"
    },
    "active": true
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c290c"
    },
    "user_id": {
      "$oid": "69fe83c33d739553f17c2908"
    },
    "type": "pin",
    "pin_hex": "BEEF",
    "reader": 1,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.544Z"
    },
    "active": true
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c290a"
    },
    "user_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "type": "card",
    "card_hex": "04A1B2C3D4",
    "card_dec": "19892716500",
    "bits": 34,
    "format": 0,
    "reader": 0,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.543Z"
    },
```

```
    "active": true
  }
]
-- findCredentialByCard("04A1B2C3D4")
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290a"
  },
  "user_id": {
    "$oid": "69fe83c33d739553f17c2907"
  },
  "type": "card",
  "card_hex": "04A1B2C3D4",
  "card_dec": "19892716500",
  "bits": 34,
  "format": 0,
  "reader": 0,
  "enrolled": {
    "$date": "2026-05-09T00:45:55.543Z"
  },
  "active": true
}
-- updateCredential(martinCardId)
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}
-- getCredentialById(martinCardId)
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290a"
  },
  "user_id": {
    "$oid": "69fe83c33d739553f17c2907"
  },
  "type": "card",
  "card_hex": "04A1B2C3D4",
  "card_dec": "19892716500",
  "bits": 37,
  "format": 0,
  "reader": 2,
  "enrolled": {
    "$date": "2026-05-09T00:45:55.543Z"
  },
  "active": true
}
-- revokeCredential(tempCredentialId)
{
  "acknowledged": true,
  "insertedId": null,
```

```
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}
-- temp credential after revoke
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290d"
  },
  "user_id": {
    "$oid": "69fe83c33d739553f17c2909"
  },
  "type": "card",
  "card_hex": "0BADF00D",
  "card_dec": "195948557",
  "bits": 32,
  "format": 1,
  "reader": 3,
  "enrolled": {
    "$date": "2026-05-09T00:45:55.545Z"
  },
  "active": false
}
-- deleteCredential(tempCredentialId)
{
  "acknowledged": true,
  "deletedCount": 1
}

=== 5. Create, read, update, delete schedules ===
-- listSchedules()
[
  {
    "_id": {
      "$oid": "69fe83c3cb35dbf571b92f28"
    },
    "name": "24/7",
    "periods": [
      {
        "days": [
          0,
          1,
          2,
          3,
          4,
          5,
          6
        ],
        "start": "00:00",
        "end": "23:59"
      }
    ]
  },
```

```
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290f"
  },
  "name": "Delete Me Schedule",
  "periods": [
    {
      "days": [
        0
      ],
      "start": "00:00",
      "end": "00:30"
    }
  ]
},
{
  "_id": {
    "$oid": "69fe83c3cb35dbf571b92f29"
  },
  "name": "Weekdays 8-18",
  "periods": [
    {
      "days": [
        0,
        1,
        2,
        3,
        4
      ],
      "start": "08:00",
      "end": "18:00"
    }
  ]
},
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290e"
  },
  "name": "Weekend Lab",
  "periods": [
    {
      "days": [
        5
      ],
      "start": "10:00",
      "end": "14:00"
    },
    {
      "days": [
        6
      ],
      "start": "11:00",
      "end": "13:00"
```

```
      }
    ]
  }
]
-- getSchedule("Weekend Lab")
{
  "_id": {
    "$oid": "69fe83c33d739553f17c290e"
  },
  "name": "Weekend Lab",
  "periods": [
    {
      "days": [
        5
      ],
      "start": "10:00",
      "end": "14:00"
    },
    {
      "days": [
        6
      ],
      "start": "11:00",
      "end": "13:00"
    }
  ]
}
-- updateSchedule(weekendLabId)
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}
-- checkSchedule("Weekdays 8-18", mondayMorning)
{
  "schedule": "Weekdays 8-18",
  "when": {
    "$date": "2026-05-11T09:15:00Z"
  },
  "allowed": true
}
-- checkSchedule("Weekdays 8-18", saturdayNight)
{
  "schedule": "Weekdays 8-18",
  "when": {
    "$date": "2026-05-09T22:00:00Z"
  },
  "allowed": false
}
-- deleteSchedule(deleteMeScheduleId)
{
```

```
  "acknowledged": true,
  "deletedCount": 1
}

=== 6. Upsert, read, list, and delete readers ===
-- upsertReader(0, ...)
{
  "acknowledged": true,
  "insertedId": {
    "$oid": "69fe83c3cb35dbf571b92f2a"
  },
  "matchedCount": 0,
  "modifiedCount": 0,
  "upsertedCount": 1
}
-- upsertReader(1, ...)
{
  "acknowledged": true,
  "insertedId": {
    "$oid": "69fe83c3cb35dbf571b92f2b"
  },
  "matchedCount": 0,
  "modifiedCount": 0,
  "upsertedCount": 1
}
-- getReader(0)
{
  "_id": {
    "$oid": "69fe83c3cb35dbf571b92f2a"
  },
  "index": 0,
  "addr": 0,
  "firmware": "2.83.0",
  "last_seen": {
    "$date": "2026-05-11T09:15:00Z"
  },
  "model": 1,
  "power": 0,
  "sc": 1,
  "serial": "21AA0145",
  "state": "ONLINE",
  "tamper": 0,
  "vendor": "E41E0A"
}
-- listReaders()
[
  {
    "_id": {
      "$oid": "69fe83c3cb35dbf571b92f2a"
    },
    "index": 0,
    "addr": 0,
    "firmware": "2.83.0",
```

```
    "last_seen": {
      "$date": "2026-05-11T09:15:00Z"
    },
    "model": 1,
    "power": 0,
    "sc": 1,
    "serial": "21AA0145",
    "state": "ONLINE",
    "tamper": 0,
    "vendor": "E41E0A"
  },
  {
    "_id": {
      "$oid": "69fe83c3cb35dbf571b92f2b"
    },
    "index": 1,
    "addr": 1,
    "firmware": "1.0.0",
    "last_seen": {
      "$date": "2026-05-09T22:00:00Z"
    },
    "model": 2,
    "power": 0,
    "sc": 0,
    "serial": "DEMO0001",
    "state": "OFFLINE",
    "tamper": 0,
    "vendor": "DEMO01"
  }
]
-- deleteReader(1)
{
  "acknowledged": true,
  "deletedCount": 1
}

=== 7. Log, query, and delete events ===
-- getEvents({ limit: 10 })
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2910"
    },
    "type": "card",
    "reader": 0,
    "hex": "04A1B2C3D4",
    "bits": 37,
    "format": 0,
    "ts": {
      "$date": "2026-05-11T09:15:00Z"
    },
    "raw": "!CARD demo",
    "logged": {
```

```
      "$date": "2026-05-09T00:45:55.567Z"
    }
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2911"
    },
    "type": "pd_status",
    "reader": 0,
    "state": "ONLINE",
    "ts": {
      "$date": "2026-05-11T09:15:00Z"
    },
    "raw": "!PD 0 ONLINE",
    "logged": {
      "$date": "2026-05-09T00:45:55.568Z"
    }
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2912"
    },
    "type": "demo_delete",
    "reader": 9,
    "ts": {
      "$date": "2026-05-09T22:00:00Z"
    },
    "raw": "!DEMO DELETE",
    "logged": {
      "$date": "2026-05-09T00:45:55.569Z"
    }
  }
]
-- deleteEvents({ type: "demo_delete" })
{
  "acknowledged": true,
  "deletedCount": 1
}

=== 8. Log, query, and delete system logs ===
-- getSystemLogs({ limit: 10 })
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2914"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.573Z"
    },
    "level": "warn",
    "source": "demo",
    "message": "Reader 1 was removed during cleanup"
  },
```

```
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2915"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.573Z"
    },
    "level": "info",
    "source": "demo-cleanup",
    "message": "This log will be deleted"
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2913"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.572Z"
    },
    "level": "info",
    "source": "demo",
    "message": "Demo script started",
    "data": {
      "step": 8
    }
  }
]
-- deleteSystemLogs({ source: "demo-cleanup" })
{
  "acknowledged": true,
  "deletedCount": 1
}

=== 9. Access evaluation helpers ===
-- evaluateUserAccess(martin, reader 0, mondayMorning)
{
  "granted": true,
  "reason": "allowed"
}
-- evaluateUserAccess(visitor, reader 0, mondayMorning)
{
  "granted": false,
  "reason": "reader not allowed"
}
-- evaluateUserAccess(visitor, reader 1, saturdayNight)
{
  "granted": false,
  "reason": "outside schedule"
}

=== 10. Access workflows and access log ===
-- accessByCard(martin card)
{
  "granted": true,
```

```
  "reason": "allowed",
  "credential": {
    "_id": {
      "$oid": "69fe83c33d739553f17c290a"
    },
    "user_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "type": "card",
    "card_hex": "04A1B2C3D4",
    "card_dec": "19892716500",
    "bits": 37,
    "format": 0,
    "reader": 2,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.543Z"
    },
    "active": true
  },
  "user": {
    "_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "username": "martin",
    "full_name": "Martin V.",
    "role": "admin",
    "active": true,
    "allowed_readers": [
      0,
      2
    ],
    "schedule": "24/7",
    "created": {
      "$date": "2026-05-09T00:45:55.531Z"
    }
  }
}
-- accessByCard(unknown card)
{
  "granted": false,
  "reason": "unknown card",
  "credential": null,
  "user": null
}
-- accessByPin(visitor pin outside schedule)
{
  "granted": false,
  "reason": "outside schedule",
  "credential": {
    "_id": {
      "$oid": "69fe83c33d739553f17c290c"
    },
    "user_id": {
```

```
      "$oid": "69fe83c33d739553f17c2908"
    },
    "type": "pin",
    "pin_hex": "BEEF",
    "reader": 1,
    "enrolled": {
      "$date": "2026-05-09T00:45:55.544Z"
    },
    "active": true
  },
  "user": {
    "_id": {
      "$oid": "69fe83c33d739553f17c2908"
    },
    "username": "visitor",
    "full_name": "Weekend Visitor",
    "role": "user",
    "active": true,
    "allowed_readers": [
      1
    ],
    "schedule": "Weekdays 8-18",
    "created": {
      "$date": "2026-05-09T00:45:55.532Z"
    }
  }
}
-- getAccessLog({ limit: 10 })
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2919"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.587Z"
    },
    "card_hex": "DEADBEEF",
    "pin_hex": null,
    "user_id": null,
    "username": null,
    "granted": false,
    "reader": 99,
    "reason": "manual demo delete"
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2918"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.586Z"
    },
    "card_hex": null,
    "pin_hex": "BEEF",
```

```
    "user_id": {
      "$oid": "69fe83c33d739553f17c2908"
    },
    "username": "visitor",
    "granted": false,
    "reader": 1,
    "reason": "outside schedule"
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2917"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.583Z"
    },
    "card_hex": "FFFFFFFF",
    "pin_hex": null,
    "user_id": null,
    "username": null,
    "granted": false,
    "reader": 0,
    "reason": "unknown card"
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2916"
    },
    "ts": {
      "$date": "2026-05-09T00:45:55.582Z"
    },
    "card_hex": "04A1B2C3D4",
    "pin_hex": null,
    "user_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "username": "martin",
    "granted": true,
    "reader": 0,
    "reason": "allowed"
  }
]
-- deleteAccessLog({ reason: "manual demo delete" })
{
  "acknowledged": true,
  "deletedCount": 1
}

=== 11. Delete temporary user and final summary ===
-- deleteUser(tempUserId)
{
  "acknowledged": true,
  "deletedCount": 1
}
```

```
-- listUsers({ activeOnly: false })
[
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2907"
    },
    "username": "martin",
    "full_name": "Martin V.",
    "role": "admin",
    "active": true,
    "allowed_readers": [
      0,
      2
    ],
    "schedule": "24/7",
    "created": {
      "$date": "2026-05-09T00:45:55.531Z"
    }
  },
  {
    "_id": {
      "$oid": "69fe83c33d739553f17c2908"
    },
    "username": "visitor",
    "full_name": "Weekend Visitor",
    "role": "user",
    "active": true,
    "allowed_readers": [
      1
    ],
    "schedule": "Weekdays 8-18",
    "created": {
      "$date": "2026-05-09T00:45:55.532Z"
    }
  }
]
-- final summary
{
  "db": "osdp_access_demo",
  "collections": [
    "access_log",
    "credentials",
    "events",
    "readers",
    "schedules",
    "system_logs",
    "users"
  ],
  "counts": {
    "users": 2,
    "credentials": 3,
    "events": 2,
    "access_log": 3,
```

```
      "readers": 1,
      "schedules": 3,
      "system_logs": 2
    }
  }
```

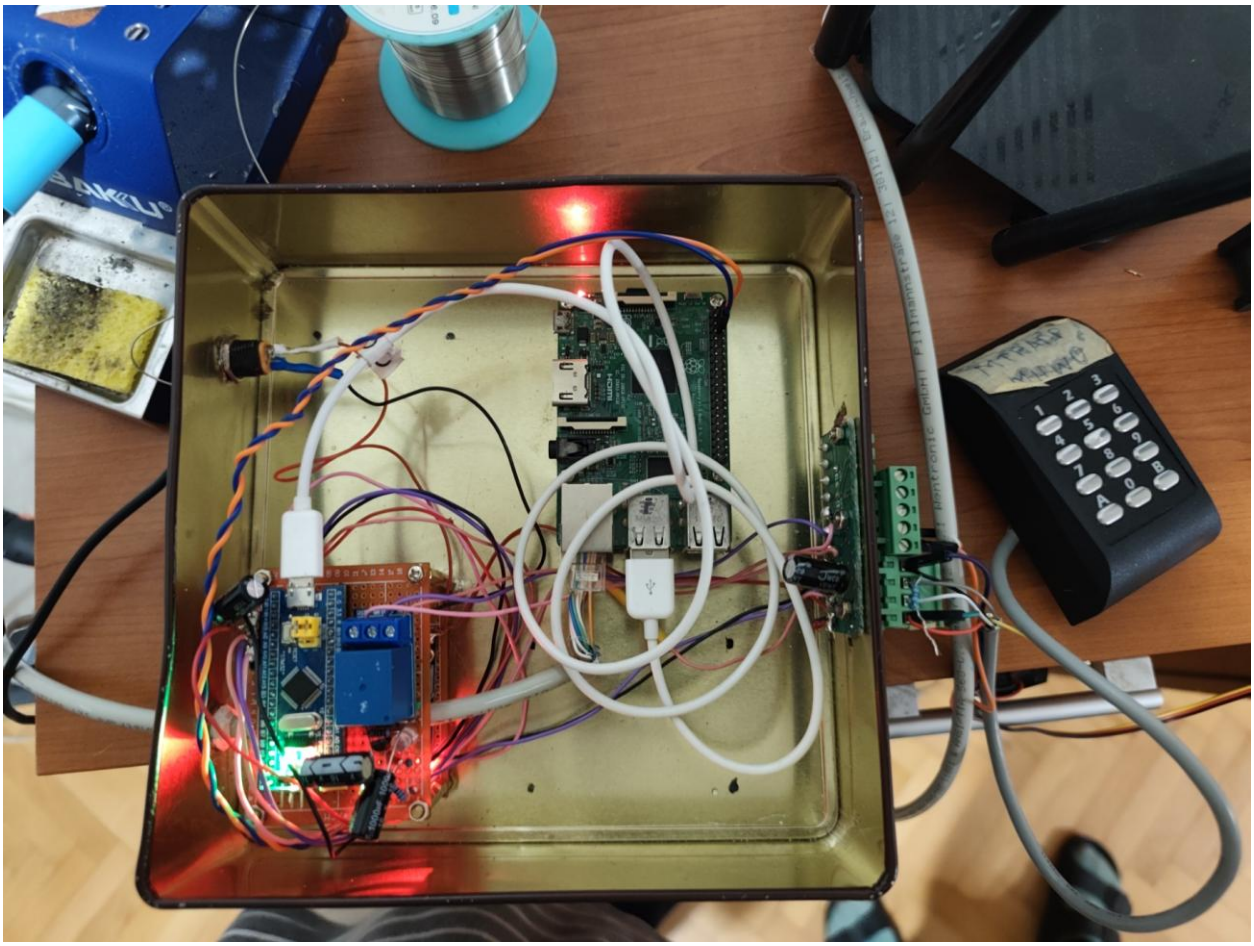

Appendix G: Prototype Hardware Photos and Web UI Walkthrough

This appendix merges the material from docs/ui-testing/ui-walkthrough.md into the main Databases II project report so the course submission has one combined Markdown source for both the database report and the visual walkthrough.

G.1 Hardware Photos

These photos show the current prototype as an open bench build rather than a finished enclosure. The visible wiring, relay board, Raspberry Pi, keypad reader, and breakout board are useful context when reading the transport, relay, and power-related discussion in the earlier chapters.

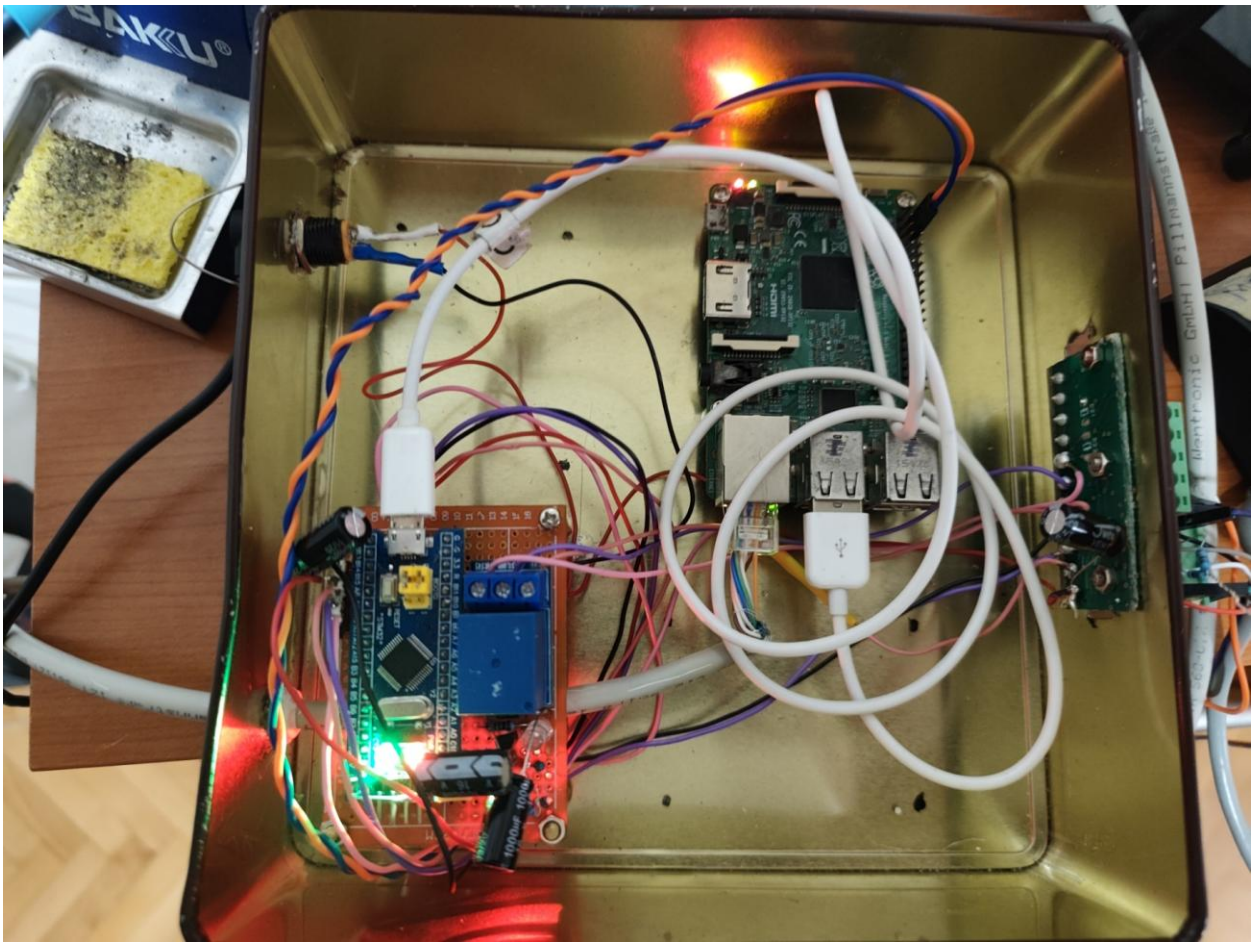
G.1.1 Hardware Overview With Keypad Attached



Hardware overview with keypad

This is the clearest top-down overview of the full test assembly. It shows the open metal enclosure, Raspberry Pi mounted inside the box, STM32 Blue Pill controller plus relay board, side terminal board, and the external keypad reader connected on the right. Use this image as the main reference for the physical test setup when discussing wiring or hardware-induced communication problems.

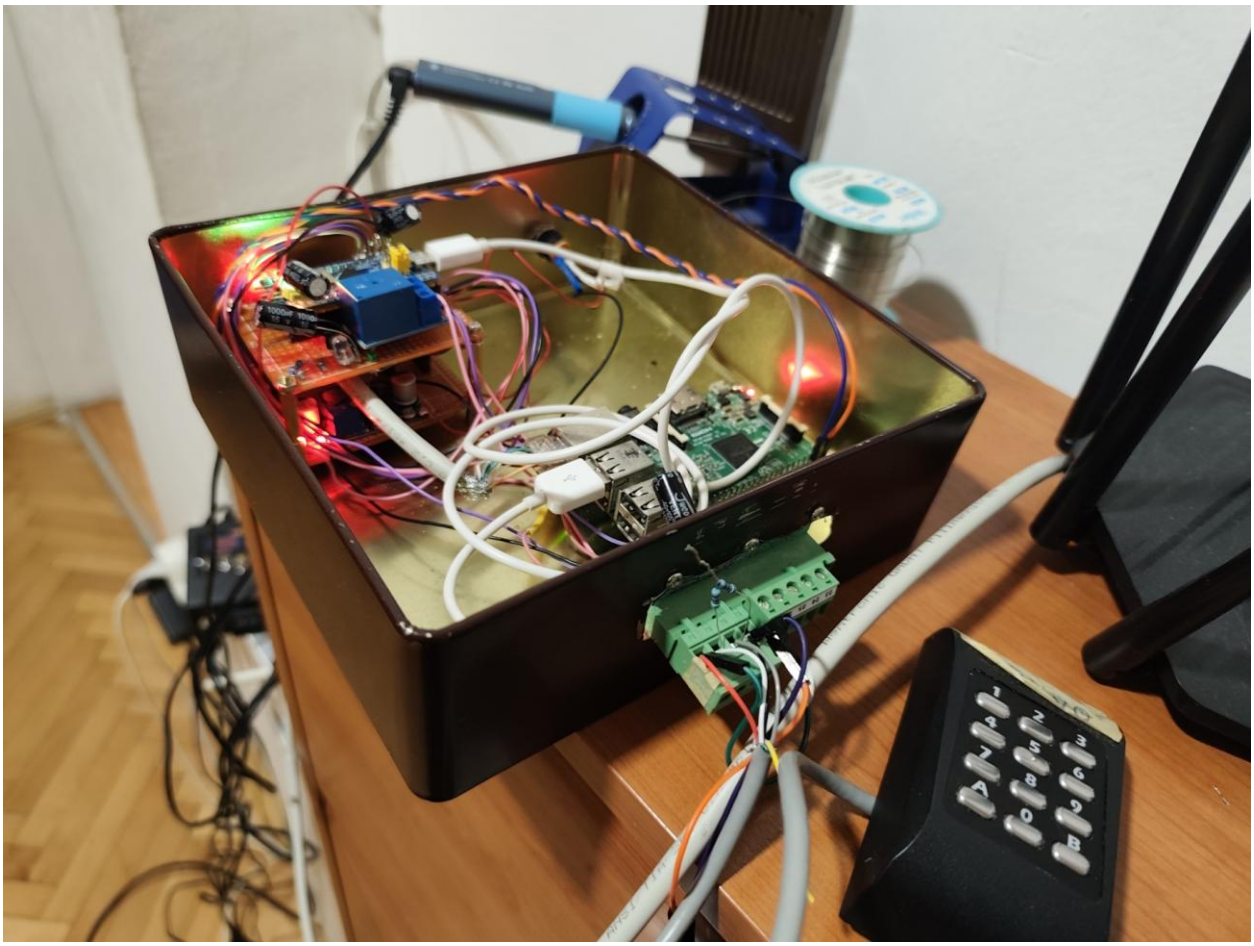
G.1.2 Internal Layout And Wiring Density



Internal layout and wiring density

This photo focuses on the inside of the enclosure and makes the wiring density more obvious. It is useful when describing why the prototype is still susceptible to bench noise, ground-routing issues, relay interference, or layout-related problems. It also shows the Raspberry Pi and controller board sharing space inside a conductive enclosure with short internal runs and improvised harnessing.

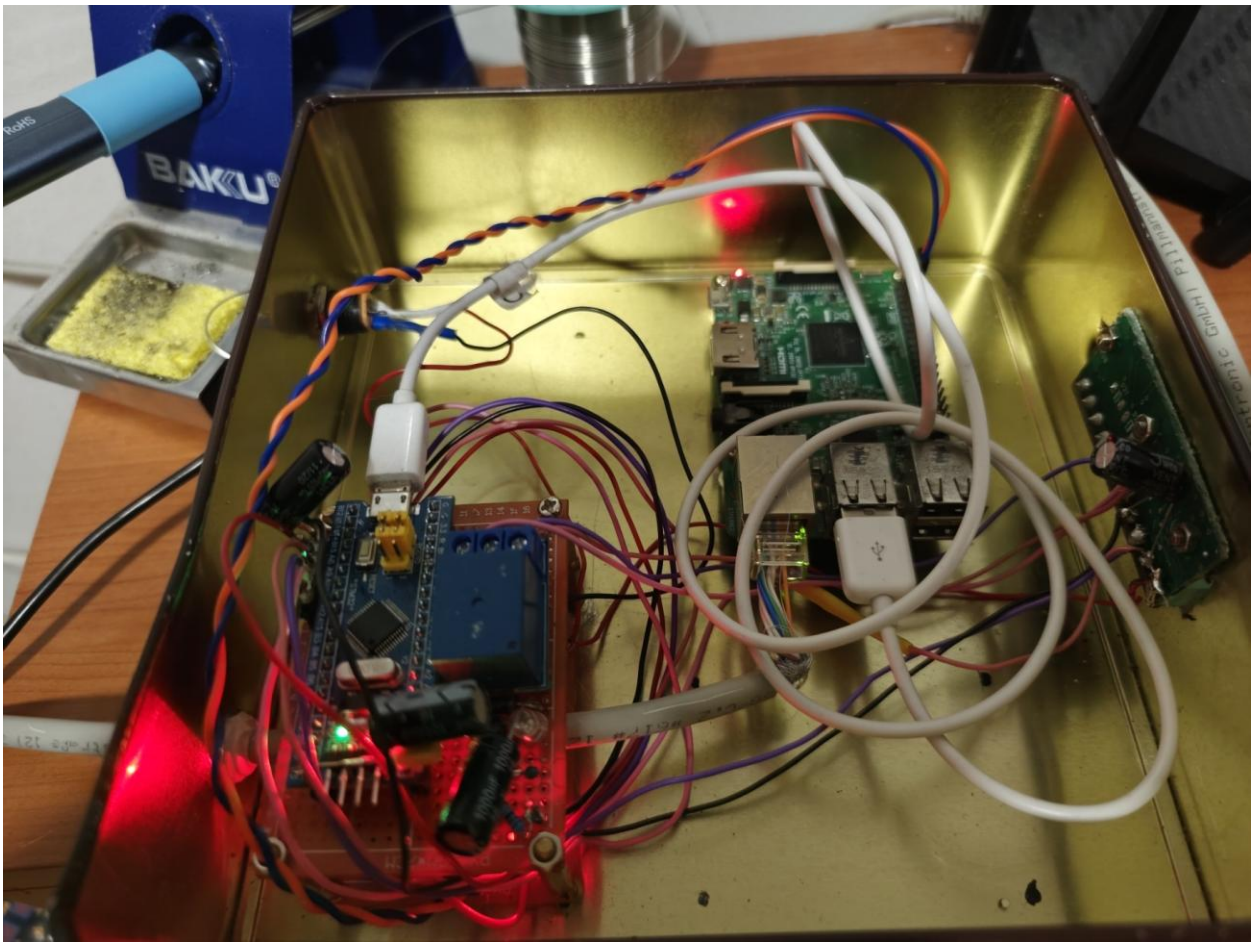
G.1.3 Angled View Of Enclosure, Terminal Block, And Keypad



Angled view of enclosure and keypad

This angle helps explain how the enclosure, external connector area, and keypad are physically arranged on the desk during testing. It is a useful image for documenting cable exits, field wiring direction, and why mechanical layout may still be part of the debugging story.

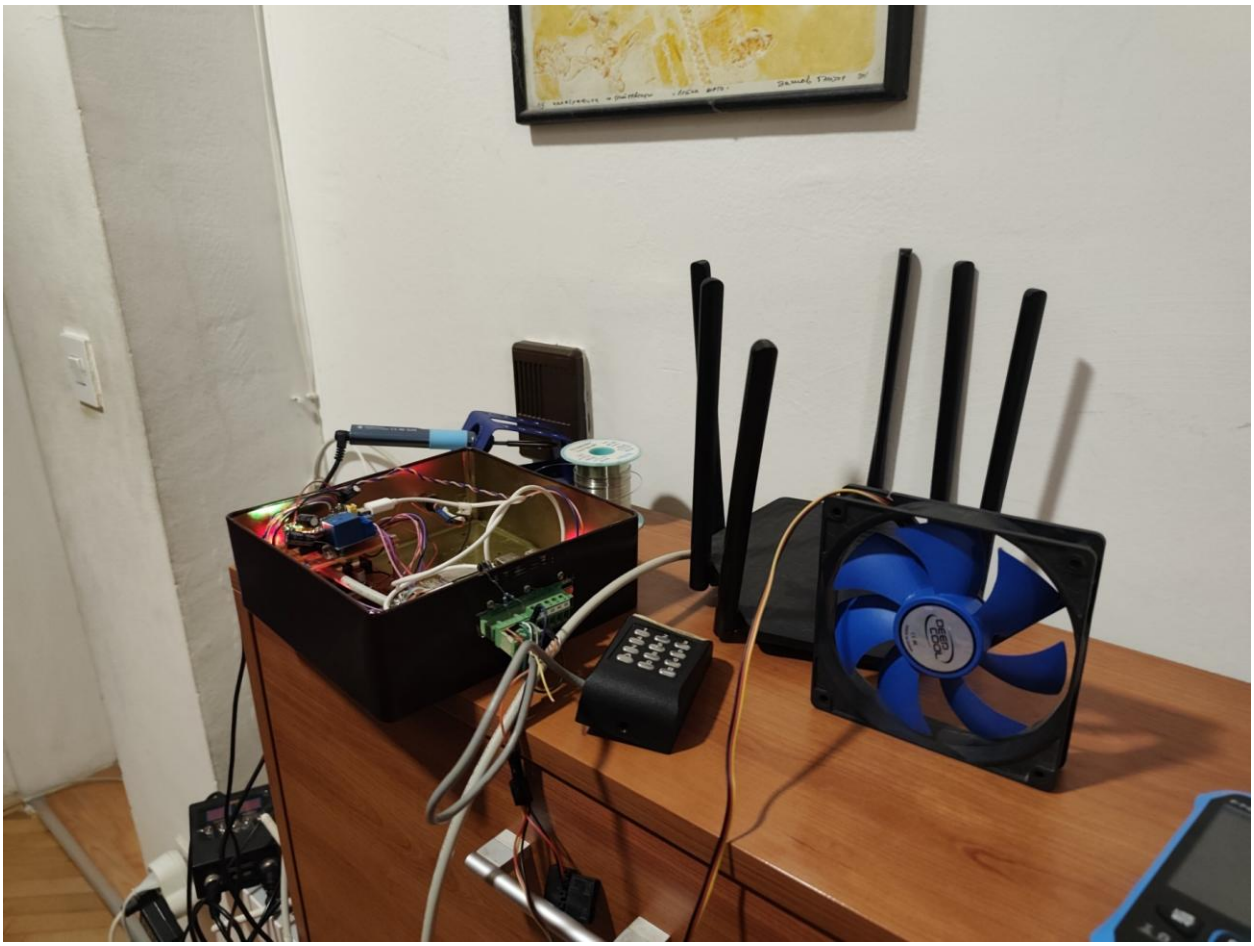
G.1.4 Close View Of Internal Components



Close internal component view

This is the most component-focused photo of the internal build. It gives a clearer view of the Blue Pill controller board, the relay module, the Raspberry Pi, and the side board used for termination or interface wiring. Use it when describing the internal relationship between the control electronics and the relay path.

G.1.5 Bench Context And Cooling Experiment



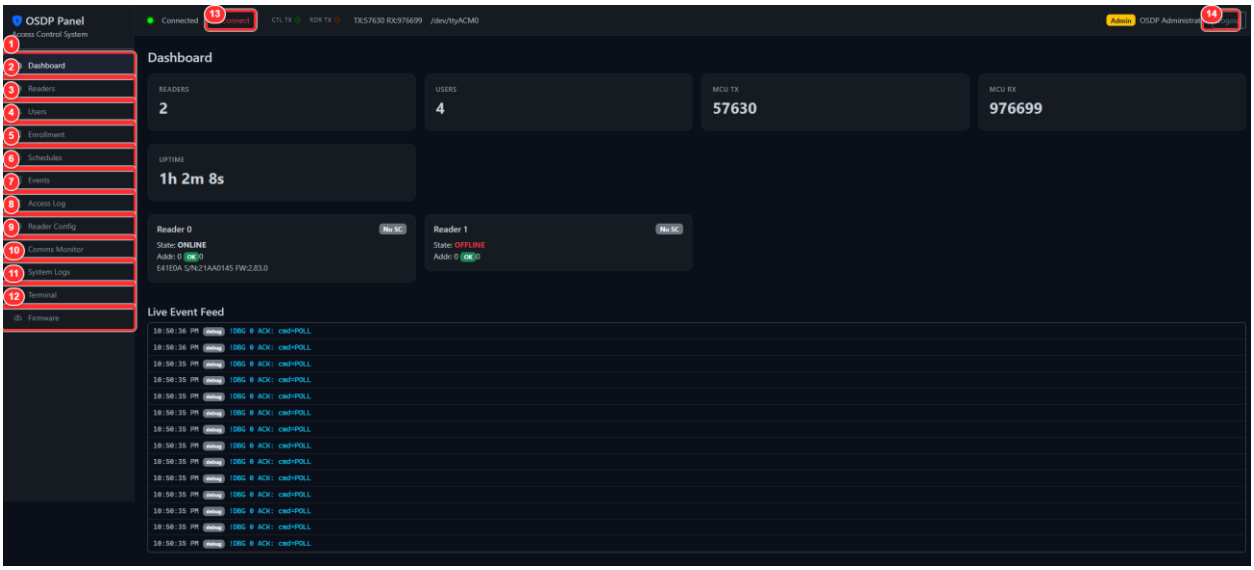
Bench context and cooling experiment

This wide shot captures the full bench context: the enclosure, keypad, nearby router, and the external fan placed beside the setup. It is useful as a documentation image because it shows that the prototype is being tested in an ad hoc lab environment rather than a fixed installation, which matters when discussing repeatability, EMI, wiring stability, and temporary cooling or isolation experiments.

G.2 Web UI Walkthrough

The following screenshots were captured from the live application at http://osdp.local:5000 while signed in as an admin user. Red squares mark the important tabs, buttons, or controls. The numbered lists under each screenshot explain what each marked item means and what it is used for during functional UI testing.

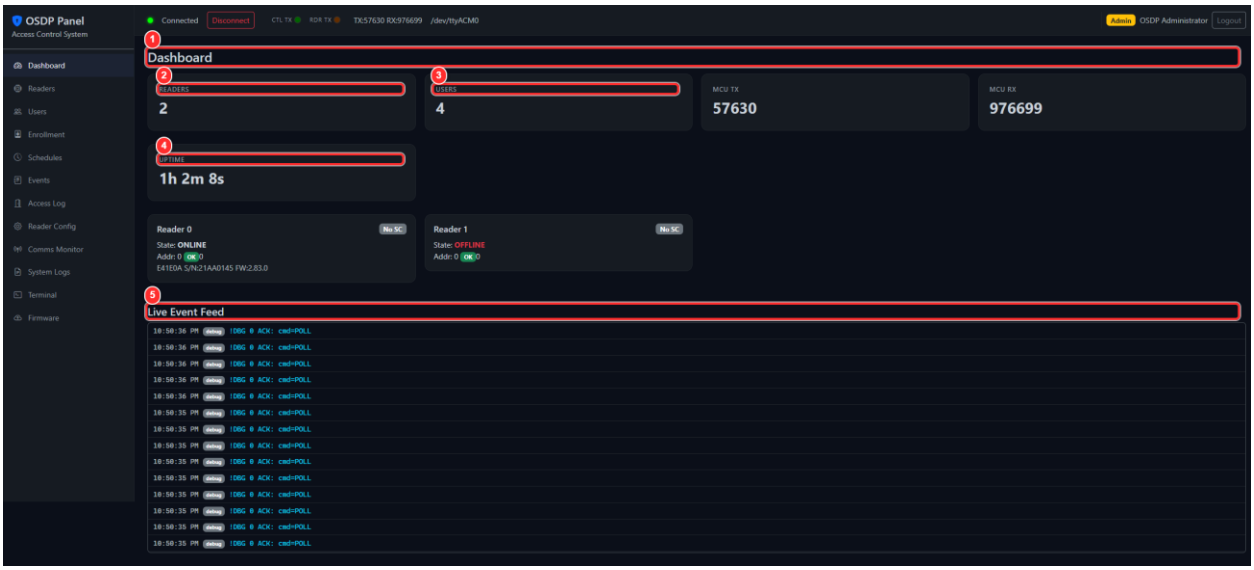
G.2.1 Overview Navigation



Overview navigation

1. Dashboard: Summary view for readers, users, controller traffic, uptime, and recent events.
2. Readers: Reader inventory and direct status or action commands.
3. Users: User records and panel login account management.
4. Enrollment: Card and PIN enrollment workflow.
5. Schedules: Time windows used by access-control rules.
6. Events: Raw event log browser.
7. Access Log: Access decision history with refresh support.
8. Reader Config: Direct configuration and command console for a selected reader.
9. Comms Monitor: Live transport and debug stream with debug toggles.
10. System Logs: Backend or system-level operational log view.
11. Terminal: Raw command entry against the controller or bridge.
12. Firmware: Firmware upload and flash workflow.
13. Connect or Disconnect: Opens or closes the bridge connection between the panel app and the controller.
14. Logout: Ends the authenticated admin session.

G.2.2 Dashboard

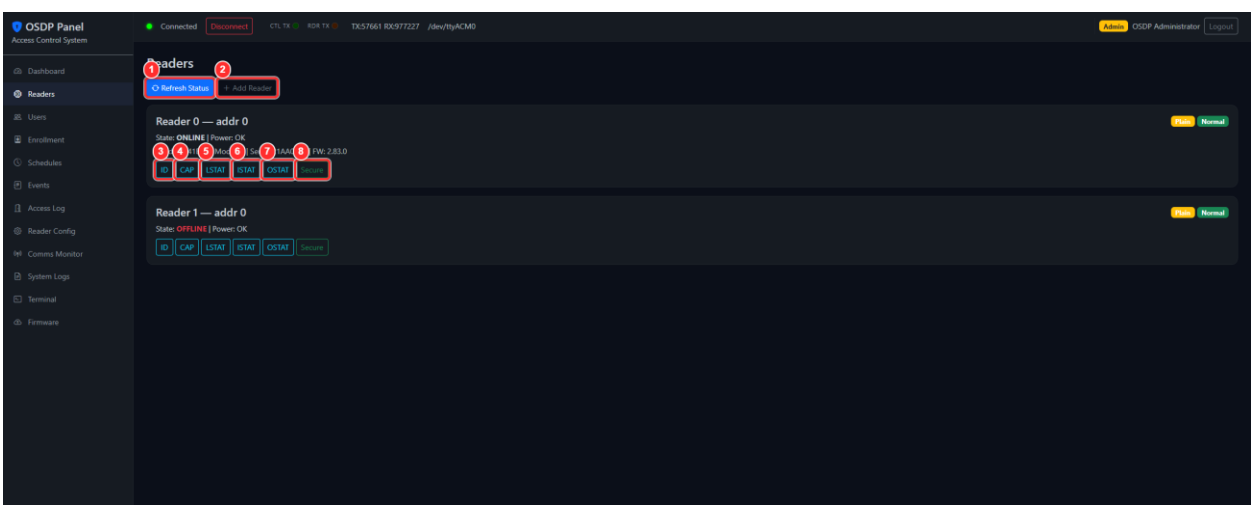


Dashboard

1. Dashboard heading: Confirms the active tab and page context.
2. Readers: High-level count of configured readers.
3. Users: High-level count of configured user records.
4. Uptime: Shows panel or backend uptime for quick health checks.
5. Live Event Feed: Real-time event stream used to validate card reads, relays, keypad traffic, and state changes.

Testing focus: confirm counts update, uptime is present, and live events move when the controller is active.

G.2.3 Readers



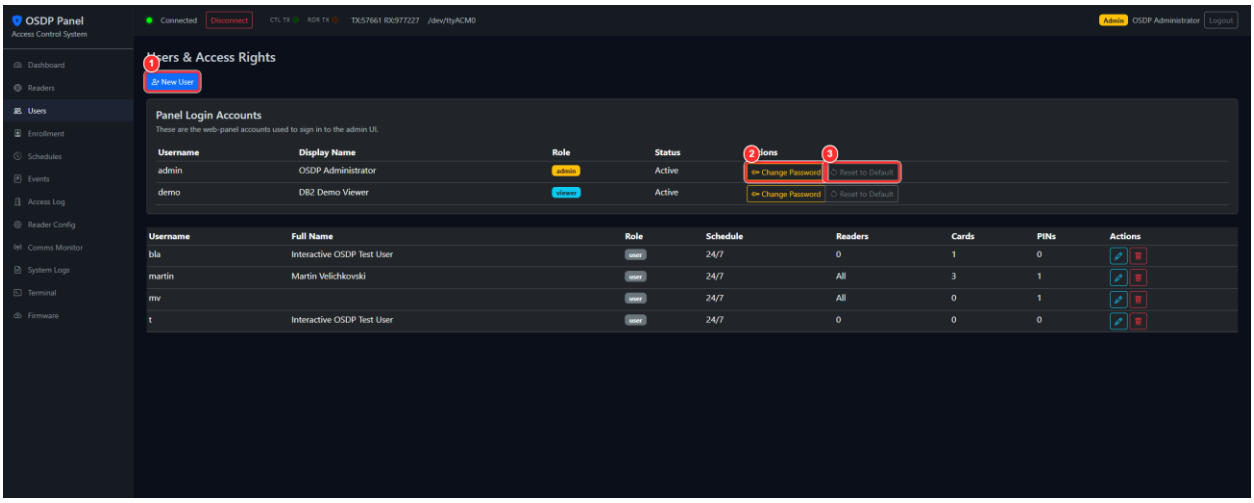
Readers

1. Refresh Status: Reloads current reader state without leaving the page.

2. Add Reader: Starts the workflow for registering another reader.
3. ID: Requests reader identity information.
4. CAP: Requests the reader capability table.
5. LSTAT: Requests local status such as tamper and power state.
6. ISTAT: Requests input status.
7. OSTAT: Requests output status.
8. Secure: Initiates secure-channel setup with the selected reader.

Testing focus: each command should produce visible feedback, status changes, or log output without hanging the page.

G.2.4 Users

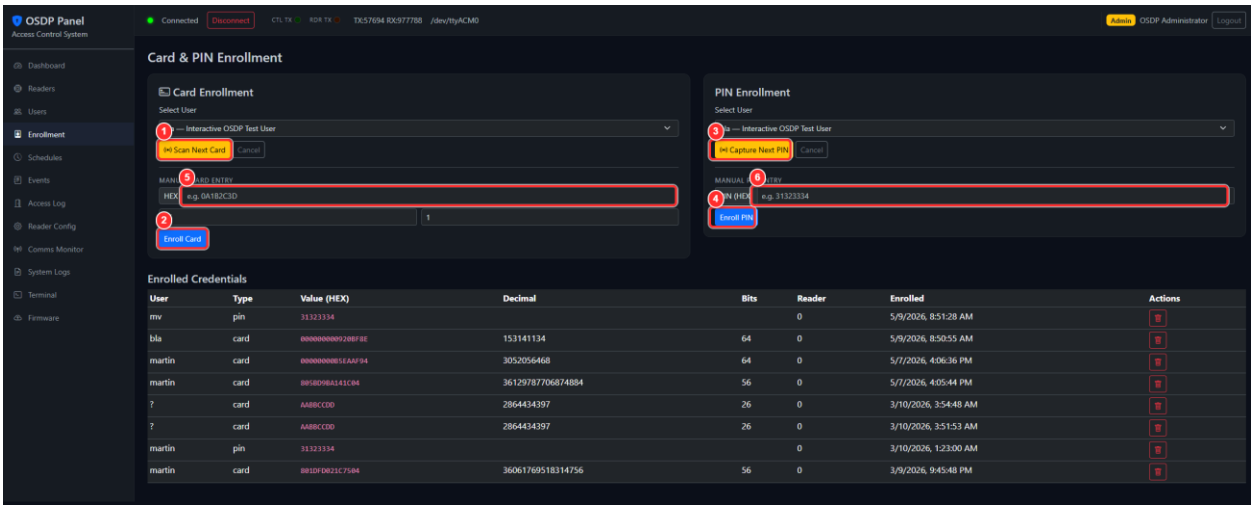


Users

1. New User: Creates a new access-control user.
2. Change Password: Changes the selected panel account password.
3. Reset to Default: Resets a panel account back to its seeded default password.

Testing focus: dialogs or forms should open correctly, validation should be clear, and account changes should persist after refresh.

G.2.5 Enrollment

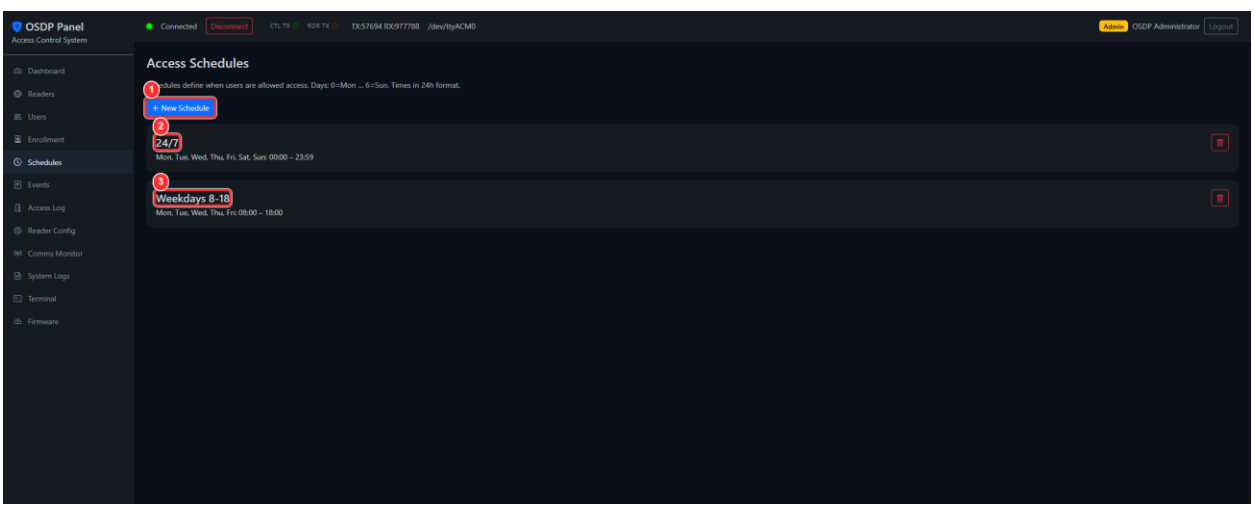


Enrollment

1. Scan Next Card: Arms the page to capture the next presented card.
2. Enroll Card: Saves the currently captured or manually entered card data.
3. Capture Next PIN: Arms the page to capture the next keypad PIN sequence.
4. Enroll PIN: Saves the currently captured or manually entered PIN data.
5. Manual card field: Used to type a card value directly when testing without a live swipe.
6. Manual PIN field: Used to type keypad data directly in hex form.

Testing focus: scan or capture should populate fields, cancel paths should reset state, and manual entry should validate format cleanly.

G.2.6 Schedules



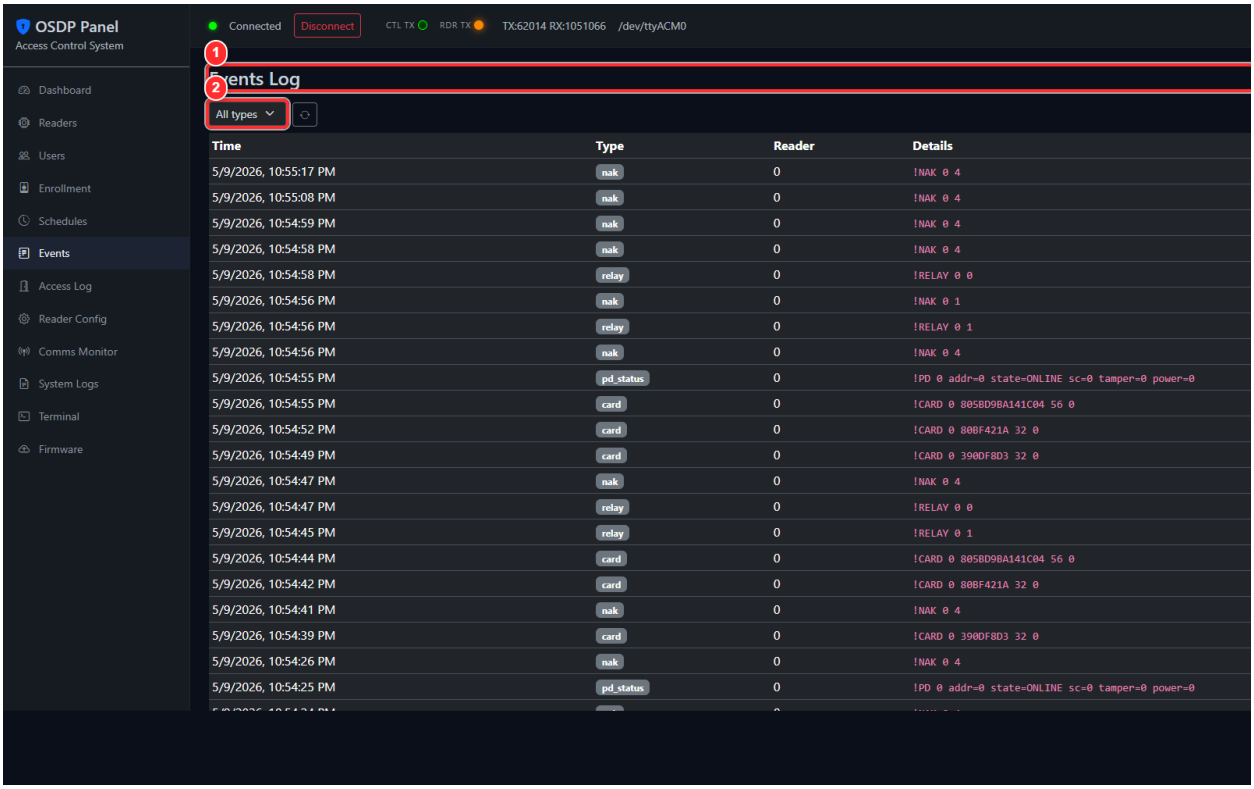
Schedules

1. New Schedule: Creates another schedule record.
2. 24/7: Example always-on schedule card.

3. Weekdays 8-18: Example limited-hours schedule card.

Testing focus: schedule cards should render correctly, edits should be discoverable, and creating a new schedule should not disturb existing entries.

G.2.7 Events

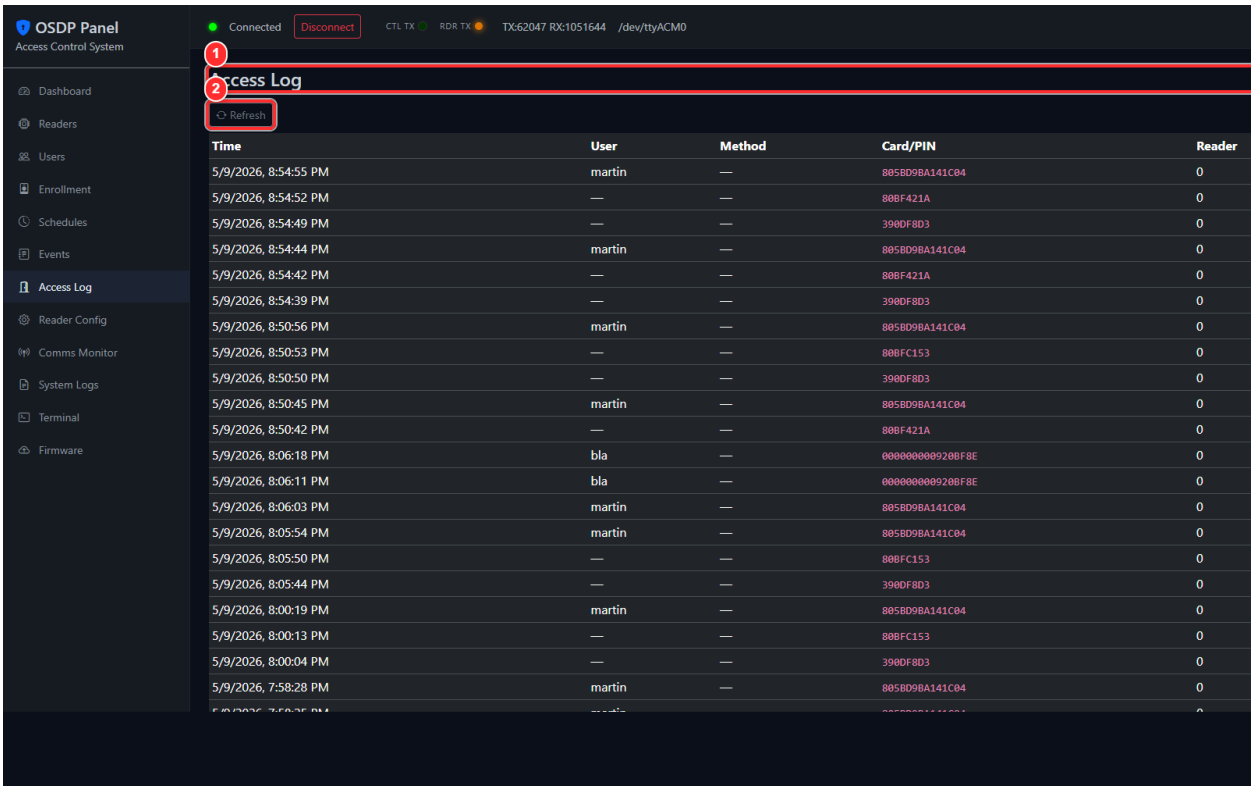


Events

1. Events Log: Main page title for the raw event history view.
2. Filter selector: Narrows the event stream by type or category.

Testing focus: filter changes should immediately affect the displayed log without breaking pagination or layout.

G.2.8 Access Log

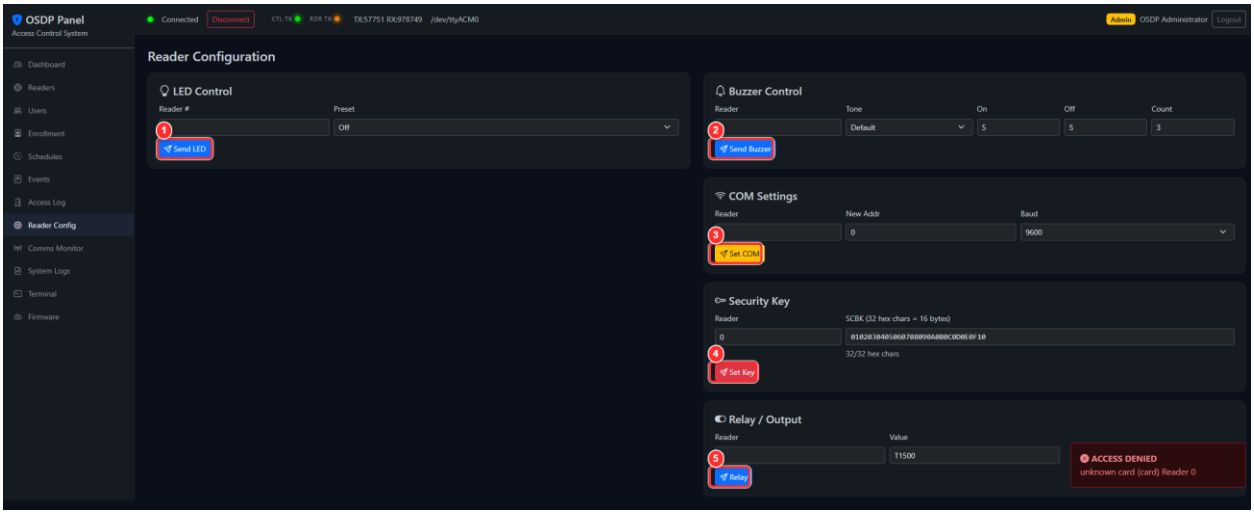


Access Log

1. Access Log: Page title for grant or deny history.
2. Refresh: Reloads the latest access decisions.

Testing focus: refresh should show new swipes and decisions, and timestamps or usernames should remain aligned after reload.

G.2.9 Reader Config

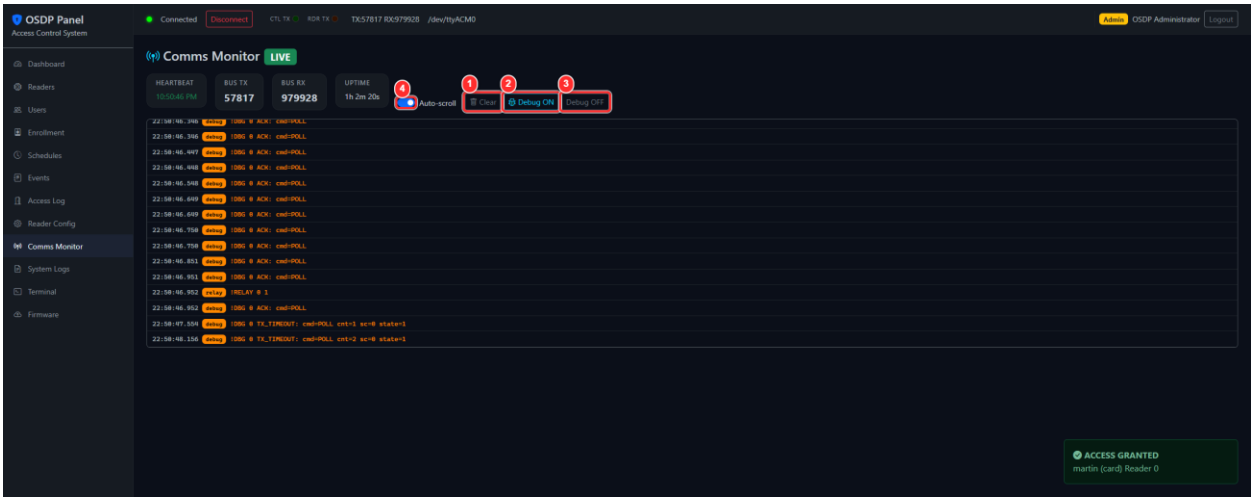


Reader Config

1. Send LED: Sends a direct LED command to the reader.
2. Send Buzzer: Sends a direct buzzer command.
3. Set COM: Pushes communication settings such as address or baud.
4. Set Key: Programs or updates the reader key material.
5. Relay: Sends a relay or output command.

Testing focus: each command should produce either a visible hardware effect or an immediate result in the comms monitor. This page is the fastest place to test direct reader actions.

G.2.10 Comms Monitor

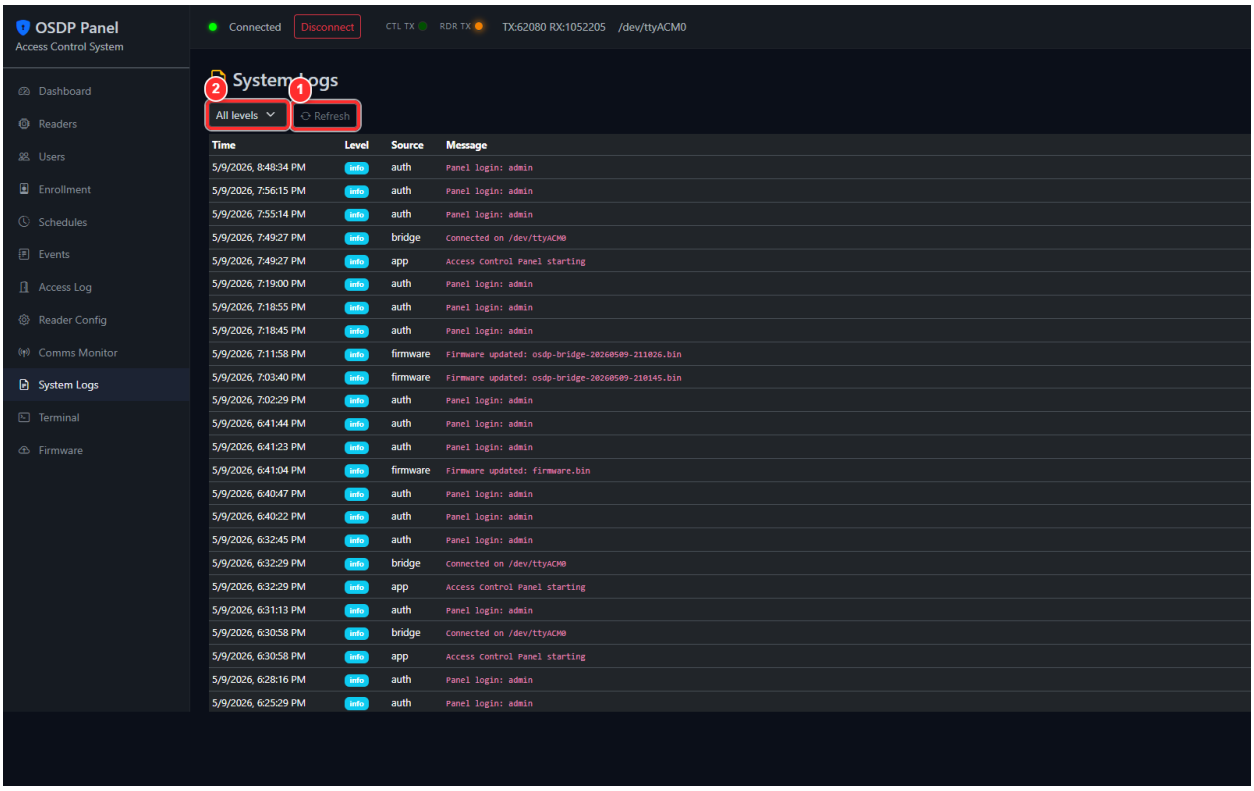


Comms Monitor

1. Clear: Clears the visible monitor output.
2. Debug ON: Enables verbose controller debug traffic.
3. Debug OFF: Disables verbose controller debug traffic.
4. Checkbox filter: Toggles an additional filtering option for the live monitor stream.

Testing focus: debug toggles should take effect immediately and the monitor should remain readable during heavy traffic.

G.2.11 System Logs



System Logs

1. Refresh: Reloads system log entries.
2. Filter selector: Narrows logs by severity or type.

Testing focus: refresh and filtering should work without duplicating entries or freezing the page.

G.2.12 Terminal

Terminal

1. Command input: Freeform raw command entry field.
2. Send: Submits the typed raw command to the controller or bridge.

Testing focus: commands should echo clearly, errors should be visible, and the terminal should not allow accidental invisible failures.

G.2.13 Firmware

Firmware

1. Choose File: Opens the file picker for a compiled firmware .bin image.
2. Flash: Starts the firmware upload once a file has been selected.
3. How it works: Reference panel that explains the bootloader and flash process.

Testing focus: invalid files should be rejected, progress should be visible during upload, and recovery instructions should remain understandable if flashing fails.

G.2.14 Used Literature

The following literature and documentation were used as technical background for the database, backend, frontend, embedded bridge, and deployment parts of the project.

Source	Used For	Reference
Security Industry Association	Open Supervised Device Protocol (OSDP).	https://www.securityindustry.org/industry-standards/open-supervised-device-protocol/
MongoDB Documentation	MongoDB Manual: data modeling, CRUD operations, and indexing.	https://www.mongodb.com/docs/manual/
MongoDB Documentation	PyMongo driver documentation: MongoClient, databases, collections, and collection operations.	https://www.mongodb.com/docs/languages/python/pymongo-driver/current/
Pallets Projects	Flask documentation for Python web application structure and routing.	https://flask.palletsprojects.com/
Flask-SocketIO	Flask-SocketIO documentation for real-time event transport in Flask applications.	https://flask-socketio.readthedocs.io/

React	React documentation for component-based frontend development.	https://react.dev/
Docker Documentation	Docker run and container documentation used for MongoDB deployment notes.	https://docs.docker.com/engine/containers/run/
Raspberry Pi Documentation	Raspberry Pi documentation for Linux-based controller host context.	https://www.raspberrypi.com/documentation/
STMicroelectronics	STM32F103C8 product and datasheet material for the STM32 Blue Pill bridge MCU.	https://www.st.com/en/microcontrollers-microprocessors/stm32f103c8.html
Analog Devices / Maxim Integrated	MAX481/MAX483/MAX485/MAX487-MAX491 RS-485/RS-422 transceiver datasheet.	https://www.analog.com/media/en/technical-documentation/data-sheets/MAX1487-MAX491.pdf